PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. under the Panjab University Act VII of 1947—enacted by the Govt. of India)

FACULTY OF ARTS
SYLLABI

FOR

M.A Economics with Research Two Year Programme
(According to NEP)

1st & 2nd Semester for Examinations 2026- 2027
and
3rd & 4th Semester Examinations 2027-2028

Semester	Level of Course	Table-1 M.A. Economics with Research			
		Course Code	**Course Name**	**Credits**	**Total Credits**
I	**L-400**	MAECODSC-101	Advanced Microeconomics-I	4	20
		MAECODSC-102	Advanced Macroeconomics-I	4	
		MAECODSC-103	Advanced Quantitative Methods for Economics-I	4	
		MAECODSC-104	International Trade and Trading Systems	4	
		MAECODSE-105(*)	Any Paper from Appendix Table-1	4	
II	**L-400**	MAECODSC-201	Advanced Microeconomics-II	4	20
		MAECODSC-202	Advanced Macroeconomics-II	4	
		MAECODSC-203	Advanced Quantitative Methods for Economics-II	4	
		MAECODSC-204	Advanced Public Finance	4	
		MAECODSE-205(*)	Any Paper from Appendix Table-2	4	
The student successfully securing 40 credits will be awarded a Post-Graduate Diploma in Economics					

	Course	Course Code	Course Name	Credits	Total Credits
III	**L-500**	MAECORBC-RM-301 (A)	Research Methodology	4	20
		MAECODSC-302	Economics of Growth, Development and Planning-I	4	
		MAECODSC-303	Indian Economy-I	4	
		MAECODSE-304(*)	Any one Papers from Appendix Table-3	4	
		MAECORBC-DST-305 (P1)	Dissertation	4	
IV	**L-500**	MAECORBC-RM-401 (B)	Research Publications and Ethics	4	20
		MAECODSC-402	Economics of Growth, Development and Planning-II	4	
		MAECODSE-404 (*)	Any One Papers from Appendix Table-4	4	
		MAECORBC-DST-405 (P2)	Dissertation	8	
The student successfully securing 80 credits will be awarded M.A./M.A. with Research in Economics					

MAECODSE-105 (B)	Labour Economics
MAECODSE-105 (C)	Optimisation Technique and Game Theory

Appendix Table-2	
MAECODSE-205 (A)	Advance Issues in Political Economy
MAECODSE-205 (B)	Environmental Economics
MAECODSE-205 (C)	Economics of Money and Banking

Appendix Table-3	
MAECODSE-304 (A)	Econometrics-I
MAECODSE-304 (B)	Economics of Agriculture-I
MAECODSE-304 (C)	Economics of Industry-I
MAECODSE-304 (D)	Mathematical Economics-I

Appendix Table-4	
MAECODSC/ DSE-403	Indian Economy-II**
MAECODSE-404 (A)	Econometrics-II
MAECODSE-404 (B)	Economics of Agriculture-II
MAECODSE-404 (C)	Economics of Industry-II
MAECODSE-404 (D)	Mathematical Economics-II

Notes:

1**.** M.A. with Research (Table-2) shall be available only to the students who score at least 60% marks in graduation.

2. ***Represents the course code for the chosen course from the appendix tables

3. **Indian Economy-II will be offered as Discipline Specific Elective only to M.A. Economics with Research students. For M.A. Economics, it will be a core paper.

4. # Evaluation as per Regulations for Awarding National Education Policy-2020 Compliant M.A. with Research by the Panjab University, Chandigarh.

5. ## Evaluation as per Regulations for Awarding National Education Policy-2020 Compliant M.A. with Research by the Panjab University, Chandigarh.

6. The students may opt for Agriculture-II / Economics of Industry-II/ Econometrics-II/ Mathematical Economics-II in Even Semester only if they have opted for Economics of Agriculture-I/ Economics of Industry-I/ Econometrics-I/ Mathematical Economics–I/etc. in an odd semester.

7. R and R indicate part I and II of dissertation, which will continue in both

SEMESTER–I

M.A./M.A. Economics with Research			
Level of Course	**Course Code**	**Course Name**	**Credits**
L-400	MAECODSC-101	Advanced Microeconomics-I	4
	MAECODSC-102	Advanced Macroeconomics-I	4
	MAECODSC-103	Advanced Quantitative Methods for Economics-I	4
	MAECODSC-104	International Trade and Trading Systems	4
	MAECODSE-105 (A) or MAECODSE-105 (B) or MAECODSE-105 (C)	Fundamental Issues in Political Economy or Labour Economics or Optimisation Techniques and Game Theory	4

MAECODSC-101: ADVANCED MICROECONOMICS-I

Topology of the code	**Discipline Specific Core**
Level of the Course	**400**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L=60 hours(4 hours aweekfor15weeks=01 semester), T=00 hours, P=0**
Credits	**4**
Max. Marks	**100Marks=Theory80marks+InternalAssessment20 marks**
Time	**3hours**

Course Objectives: The objective of the paper is to rigorously and comprehensively equip the students with theoretical concepts, methodology and process of reasoning involved in analysing economic behaviour of individuals, firms and markets using, in general, a static and partial equilibrium framework.

Pedagogy of the Course Work: The time–tested method of imparting verbal instructions through lectures would be used. Examples, in sofar as possible, would be selected from everyday life/experience. The feedback would be sought throughthe students' written assignments, tutorials and seminars.The students would be evaluated at the end of the academic year through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Course outcomes: On successful completion of this course, learners will acquire a rigorous understanding of advanced microeconomic theory and methodology to analyse the behaviour of consumers, firms, and markets. They will be able to apply equilibrium concepts, consumer choice theory under certainty and uncertainty, and models of risk and information asymmetry to evaluate real-world economic and policy issues. Learners will develop analytical and problem-solving skills to derive demand and cost functions, assess welfare implications, and examine production decisions, returns to scale, and cost dynamics, including economies of scale and scope. The course enables learners to think critically, use formal reasoning, and apply microeconomic tools effectively in academic research, policy analysis, and professional economic practice.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:

1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question would be compulsory, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks (10x2=20).
4. Rest of the paper shall contain 4 units. Each Unit shall have two questions and the candidates shall be required to attempt on question from each Unit – 4 in all. Each question shall carry 15 marks (15 x 4 = 60).
5. Candidates are allowed to use Simple, Non-Scientific and Non-Programmable Calculator in the Examination Hall.
6. It is compulsory for the paper setter to mention the requirement of log, antilog and statistical tables.

UNIT-I

Central Ideas of Economics: How do People Make Decisions? People: face 'Trade Offs'; Incur Opportunity Costs and Gain Benefits of Decisions; Think at the Margin and Respond to Incentives. How do People Interact? Specialization and Exchange can make each party 'Better Off' Market often work efficiently, The Invisible Hand; Three Coordination Tasks in the Economy: Which Commodity be Produced and How much of it? Who should produce it and how? Who should Consume and How much of it?

Methodology of Economics as a Social Science: Alternative Processes of Construction of Positive Economic Theories.

Equilibrium: Concept, Stability, Existence and Uniqueness of Equilibrium.

Types: Static and Dynamic Equilibrium. Partial and General Equilibrium. Stability Analysis: Marshallian and Walrasian Approaches.

UNIT- II

Analysis of Consumer Choice under Certainty: Cardinal Analysis, Ordinal Utility Approach; Revisiting Consumer Equilibrium and Effects of Changes in Relative Prices and Money Incomes and Derivation of 'Marshallian' and 'Compensated' Demand Curves.

Consumer Surplus: Hicks' and Marshall's Approach.

Application of Indifference Curves: Cash Subsidy vs. Kind Subsidy: Income Leisure Choice; Intertemporal Choice of Consumption.
Revealed Preference Theory and Consistency of Choice.

UNIT- III

Market Demand: The Role of Network Externalities; 'Bandwagon', 'Veblen' and 'Snob' Effects.

Analysis of Consumer Choice under Uncertainty: Measurement of Risk. Expected Utility and Preferences towards Risk. Why do people gamble (insure)? Risk Return Trade Off. Reducing Risk.

Analysis of Consumer Behavior under Asymmetric Information:
Hidden Actions: Principals, Agents and Moral Hazard.
Hidden Characteristics: Adverse Selection Problem. Market and Government Responses to the Problems of Moral Hazard and Adverse Selection.

UNIT-IV

Theory of Production and Costs: Production, Input Choice and Cost with One Variable Input. Law of Returns to Variable Proportions and Choice of Optimum Quantity of an Input. Derivation of Short Run Cost Curves from Productivity Curves of a Variable Factor of Production. Reserve Capacity and the Shape of Short Run Cost Curves.

Multiple Input Decisions: The Choice of Optimal Techniques of Production and Factor Combinations. Factor Price Changes: Substitution and Output Effects. Elasticity of Input Substitution. Constant, Increasing and Decreasing Returns to Scale. Expansion Path and derivation of Long Run Cost of Production Curves.

Economies and Diseconomies of Scale: Shape of Short and Long Run Costs Curves: Theory versus Empirical Evidence. Multi Product Firms and Dynamic Changes in Costs: Economies of Scope; The Learning Curve Analysis.

Recommended Readings:

1. Maddala, G. S. & Miler, E. (2004). Micro-Economic Theory and Applications, Tata McGraw Hill.
2. Mankiw, G. N. (2017). Principles of Economics (3rd ed.). Thomson.
3. Pindyck, Robert, S., Rubinfel, D. L. & Gupta, P.L. (2017). Micro-Economics. New Delhi: Prentice Hall of India.
4. Boumol, W. J. & Blinder, A. S. (2007). Macroeconomics: Principles and Policy (9^{th}ed.). Thomson.
5. Briet, W. & Hochman, H. M. (1977). Readings in Micro Economics. Dryden Press.

6. Ferguson, C. E. & Gould, J. P. (1989). Micro Economic Theory (6th ed.). All India Book Seller.
7. Koutsoyiannis, A. (2016). Modern Micro Economics. London: McMillan Press.
8. Ryan, W. J. L. & Pearce, D. W. (1989). Price Theory. New Delhi: McMillan India.
9. Salvatore, D. (2007). Microeconomics: Theory and Applications, (4th ed.). Oxford University Press.

MAECODSC-102: ADVANCED MACROECONOMICS – I

Topology of the code	**Discipline Specific Core**
Level of the Course	**400**
Course Duration/LTP (Lectures, Tutorials, Practical)	**L=60 hours(4 hours aweekfor15weeks=01 semester), T=00 hours, P=0**
Credits	**4**
Max. Marks	**100Marks=Theory80marks+InternalAssessment20 marks**
Time	**3hours**

Objectives: Macroeconomics or aggregative economics analysis establishes the functional relationship between the large aggregates. The aggregate analysis has assumed such a great significance in recent times that a prior understanding of macroeconomic theoretical structure is considered essential for the proper comprehension of the different issues and policies. Macroeconomics now is not only a scientific method of analysis; but a body of empirical economic knowledge. The paper entitled "Macro-Economics-I" equips the students at the postgraduate level to understand systemic facts and latest theoretical developments for empirical analysis.

Pedagogy of the Course Work: The time–tested method of imparting verbal instructions through lectures would be used. Examples, in sofar as possible, would be selected from everyday life/experience. The feedback would be sought throughthe students' written assignments, tutorials and seminars.The students would be evaluated at the end of the academic year through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Course outcomes: Upon completion of this course, students will be able to explain and analyze the determination of income, output, and employment using classical, Keynesian, and modern macroeconomic theories. They will understand the functioning and interrelationship of commodity, labour, money, and bond markets, and assess the role of wage–price flexibility. Students will be able to compare major theories of consumption, saving, and investment under certainty and uncertainty. The course will also enable students to evaluate money supply and demand theories and apply macroeconomic reasoning to policy and real-world economic issues.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:

1. The syllabus of this paper has been divided into four units.

2. There shall be 9 questions in all.
3. The first question would be compulsory, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks (10x2=20).
4. Rest of the paper shall contain 4 units. Each Unit shall have two questions and the candidates shall be required to attempt on question from each Unit – 4 in all. Each question shall carry 15 marks (15 x 4 = 60).
5. Candidates are allowed to use Simple, Non-Scientific and Non-Programmable Calculator in the Examination Hall.
6. It is compulsory for the paper setter to mention the requirement of log, antilog and statistical tables.

UNIT-I

Fundamental Theories of Income and Employment Determination

Income and Employment Determination: Integrated Classical and Keynesian Models of Income and Employment Determination; commodity, money (including bond market of Keynes), and labour markets.

Wage-Price Flexibility and Automatic Full Employment: Classical Versus Keynesian Approach.

UNIT-II

Consumption Theories

Consumption and Consumption Function: Keynes Consumption and saving functions under psychological law of consumption, Consumption Puzzle: Absolute Income hypothesis, Relative Income hypothesis, Permanent Income hypothesis and Life Cycle Hypothesis.

Consumption under Uncertainty: Random Walk Hypothesis; Interest Rate and Saving; Consumption and Risky Asset: Consumption CAPM.

UNIT-III

Investment Theories

Investment and Investment Function: Type of Investment, Role of investment using Investment Multiplier, Classical and Keynesian Theories of Investment, Accelerator Theory of Investment, NeoClassical Theory of Investment and Tobin's-q Theory of Investment. Effects of Uncertainty, Kinked and Fixed Adjustment Costs, Investment in the Housing Market.

UNIT-IV

Money Supply and Demand

Supply of Money: Theoretical Debate and Empirical Attempts to define money; Components of Supply of Money, Credit Creation by Commercial Banks, Money Multiplier.

Demand for Money: Classical Quantity Theory, Keynesian Theory, Baumol and Tobin's Contributions. Friedman's Restatement of Quantity Theory of Money.

Recommended Readings:

1. Ackley, G. (2007). Macroeconomic Theory. Macmillan.
2. Allen, R.G.D. (1967). Macro-Economic Theory: A Mathematical Treatment. Palgrave Macmillan.
3. Blanchard, O. (2009). Macroeconomics. New Delhi: Pearson Education Inc.
4. Blanchard, O.J. & Fischer. (1996). Lectures on Macroeconomics, Delhi: PHI Learning Pvt. Ltd.
5. Branson, W. H. (1989). Macro Economic Theory and Policy (3rd ed.). Harper Collins.
6. D'Souza, E. (2008). Macroeconomics. Pearson Education.
7. Dornbusch, R., Fisher, S & Startz, R. (2002). Macroeconomics, (9th ed.). New Delhi: Tata McGraw Hills.
8. Levacic, R. & Rebmann, A. (1982). Macroeconomics: An Introduction to Keynesian Neoclassical Controversies. Macmillan.
9. Miller, R. L.& Pulsinelli, R. (1986). Macroeconomics. New York: Harper & Row.
10. Ola Olsson (2010). Essentials of Advanced Macroeconomic Theory, Routledge.
11. Romer, D. (1996). Advanced Macroeconomics. McGraw-Hill.
12. Abel, A. B., Bernanke, B. S. (2007). Macroeconomics. New Delhi: Pearson Education Inc.
13. Dornbusch, R. (1980). Open Economy Macroeconomics. New York: Weltwirtschaftliches Archive.
14. Frisch, H. (1983). Theories of Inflation. Cambridge: Cambridge University Press.
15. Froyen, R. T. (2002). Macro Economics. New Delhi: Pearson Education Inc.
16. Gordon, R. J. (2012). Macroeconomics. Delhi: PHI Learning Pvt. Ltd.
17. Heijdra, B. J. (2017). The Foundations of Modern Macroeconomics. Oxford University Press.
18. Hicks, J. R. (1950). A Contribution to the Theory of Trade Cycle Oxford: Clarendon Press.
19. Lucas, R. (1981). Studies in Business Cycle Theory. , Massachusetts: MIT Press Cambridge.
20. Makin, A. J. (2002). International Macroeconomics. Princeton Hall.
21. Mankiw, N. G. & Romer, D. (1991). New Keynesian Economics. MIT, 2 Volumes.
22. Mathews, R. C. O. (1964). The Trade Cycle. Cambridge: Cambridge University Press.
23. Pentecost, E. (2000). Macroeconomics – An Open Economy Approach. MacMillan Press Ltd.
24. Raghbendra, J. (1964). Contemporary Macroeconomics: Theory and Policy. New Age International Publishers.
25. Sherffrin, S. (1999). Rational Expectations. Cambridge University Press.
26. Snowden, B & Vane, H. R. (2005). Modern Macroeconomics: Its Origin, Development and Current Stat., UK: Edward Press.
27. Wendy, Carlin/Soskice, D. (2007). Macroeconomics: Imperfections, Institutions and Policies. Oxford University Press.

MAECODSC-103: ADVANCED QUANTITATIVE METHODS FOR ECONOMICS – I

Topology of the code	**Discipline Specific Core**
Level of the Course	**400**
Course Duration/LTP (Lectures, Tutorials, Practical)	**L=60 hours(4 hours aweekfor15weeks=01 semester), T=00 hours, P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment20 marks**
Time	**3hours**

Objectives:

The objective of this paper is to train the students in the use of mathematical and statistical tools in analysing economic problems. The course content contains simple tools and techniques, thought necessary for data collection, presentation, analysis and drawing inferences about various statistical hypotheses.

Pedagogy of the Course Work:

The time–tested method of imparting verbal instructions through lectures would be used. Examples, in so far as possible, would be selected from everyday life/experience. The feedback would be sought through the students' written assignments, tutorials and seminars. The students would be evaluated at the end of the academic year through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Course outcomes:

After completing this course, students will be able to apply differential calculus and optimization techniques to analyse economic relationships such as elasticity, cost, and revenue functions. They will develop competency in matrix algebra and growth models for solving systems of equations and evaluating economic dynamics. Learners will be able to use correlation, regression, and time series methods to interpret economic data and identify trends and seasonal patterns. The course also equips students with the ability to construct and evaluate index numbers and apply probability theory to economic decision-making. Overall, students will gain strong analytical and quantitative skills essential for economic analysis and empirical research.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:

1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question would be compulsory, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks (10x2=20).
4. Rest of the paper shall contain 4 units. Each Unit shall have two questions and the candidates shall be required to attempt on question from each Unit – 4 in all. Each question shall carry 15 marks (15 x 4 = 60).

5. Candidates are allowed to use Simple, Non-Scientific and Non-Programmable Calculator in the Examination Hall.
6. It is compulsory for the paper setter to mention the requirement of log, antilog and statistical tables.

UNIT – I

Differentiation: Review of Simple differentiation; Partial and total derivatives; Homogenous function; Euler's theorem, Maxima and Minima of a function of two independent variables; Constrained optimization; Lagrange's method.
Economic Applications: Elasticity of demand; Partial elasticities; Competitive and Complementary Commodities; Optimization Problems in cost and Revenue.

UNIT – II

Matrices: Review of adjoint and inverse of matrices; Rank of Matrix; Linear Independence and Dependence of Vectors/Matrices; Solutions of Equations (up to three) by Matrix Inverse and Gauss elimination Methods; Consistency of Equations; Quadratic Forms – Types and Properties. Simple Economic Applications.

UNIT – III

Correlation and Regression Analysis: Multiple and Partial Correlation involving three Variables. The Linear Regression Model; Finding the Multiple Linear Regression Equation and Coefficients up to two explanatory variables: Interpretation of the Coefficients.

Time Series Analysis: Definition and components. Methods of Trend Measurement: Quadratic, Exponential and Modified Exponential. Seasonal Indices by Methods of Simple Average, Ratio to Moving Average and Ratio to Trend.

UNIT – IV

Index Numbers: Concepts of Price Relative, Quantity Relative and Value Relative. Laspeyer's, Paasche's and Fisher Index Numbers. Test of an Ideal Index Number. Problems in the Construction and Limitation of Index Numbers.

Probability Theory: Classical, empirical and Axiomatic Definitions of Probability. Laws of Addition and Multiplication (with examples); Concept of Random Variable; probability mass and density functions; Expected Values.

Recommended Readings:

Allen, R. G. D. (latest edition). *Mathematica*l *Analysis for Economists*. New Delhi: Trinity Press.

Allen, R. G. D. (1986). *Mathematica*l *Analysis for Economists.* Macmillan Publisher India Ltd.

Arora, P. N., Arora, S., Arora, S. & Arora, A. (2007). *Comprehensive Statistical Methods.* New Delhi: S. Chand & Company Ltd.

Chiang, A. C. &Wainwright, K. (2005). *Fundamental Methods of Mathematical Economics* (4th ed.). New York: McGraw Hill.

Devore, J. L. (2012). *Probability and Statistics for Engineering and the Sciences*, (8th ed.), Cengage Learning India.

Freund, J. E. (1992). *Mathematical Statistics*. (5th ed.). New Delhi: Prentice-Hall, Cengage Learning, India.

Gupta, S. C. (latest edition). *Fundamentals of Statistics* (7th ed.). Himalaya Publications.

Gupta, S. P. (2014). *Statistical Methods.* New Delhi: Sultan Chand & Sons.

Kandoi, B. (2012). *Mathematics for Business and Economics with Applications*, Vol. 1. Mumbai: Himalaya.

Simon, C. P. & Lawrence, B. (2006). *Mathematics for Economists* (Indian ed.). Viva –Norton Student edition: Norton and Company.

Spiegel, M. R. & Stephens L. J. (2000). *Theory and Problems of Statistics:* Schaum's Series, *(*3rd ed.).New Delhi: Tata McGraw Hill.

Weber, J. E. (1982). *Mathematical Analysis (Business and Economic Applications)* (4th ed.).

New York: Harper and Row.

Yamane, T. (2006). *Mathematics for Economist-an Elementary survey* (2nd ed.). New Delhi: Prentice Hall.

Yamane, T. (1972). *Fundamental Methods for Mathematical Economics*. New Delhi: Prentice-Hall.

MAECODSC-104: INTERNATIONAL TRADE AND TRADING SYSTEMS

Topologyofthecode	**Discipline Specific Core**
LeveloftheCourse	**400**
CourseDuration/LTP (Lectures, Tutorials, Practical)	**L=60 hours(4 hours aweekfor15weeks=01 semester), T=00 hours, P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment20 marks**
Time	**3hours**

Objectives:
The Course intends to provide a deep understanding about the broad principles and environment, which govern the free flow of trade in goods, services and capital – both shortterm and long-term – at the global level. The contents of the paper spread over different modules, lay stress on the theory and natureof the subject which, in turn, will greatly help them to examine the impact of the trade policies followed both at the national and international levels as also their welfare implications at macro level and the distribution of gains from trade. The time-tested method of imparting verbal instructions through lectures would be used. Examples, in so far as possible, would be selected from everyday life/experience. The feedback would be sought through the students writing assignments, tutorials and seminars.

Pedagogy of the Course Work:
The time–tested method of imparting verbal instructions through lectures would be used. Examples, in sofar as possible, would be selected from everyday life/experience. The feedback would be sought throughthe students' written assignments, tutorials and seminars.The students would be evaluated at the end of the academic year through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Course outcomes:
Upon completion of this course, students will be able to explain and critically evaluate classical and modern theories of international trade under perfect and imperfect competition, including intra-industry trade and dumping. They will understand the patterns, causes, and welfare effects of international factor movements, including labour and capital mobility. Students will be able to analyze the evolution and functioning of the international monetary system and institutions such as the IMF, World Bank, and ADB. The course will also enable students to assess the structure, functions, and agreements of the WTO, including multilateral trade agreements and dispute settlement mechanisms.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:
1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question would be **compulsory** and shall be a short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be

required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks (10×2 = 20).

4. The rest of the paper shall contain four units. Each unit shall have two questions and the candidates shall be required to attempt one question from each unit – 4 in all. Each question shall carry 15 marks (15×4 = 60).
5. Candidates are allowed to use Simple, Non-Scientific and Non-Programmable Calculator in the Examination Hall.
6. It is compulsory for the paper setter to mention the requirement of log, antilog and statistical tables.

UNIT – I
International Trade Theory

Trade Based on Absolute Advantage: (Adam Smith), Comparative Advantage (David Ricardo) Advantage and Opportunity Costs (Haberler's theory, Gains from trade under constant cost as well as increasing costs).

Resources and Trade: Heckscher-Ohlin Model, Leontief Paradox.

Imperfect Competition and International Trade (Intra-industry trade), Trade Based on Dynamic Technological Differences (Technological Gap and Product Cycle Models), Gravity Model of Trade.

The Standard Trade Model: Offer Curves, Terms of Trade, Trade and Economic Growth: Immiserating Growth

UNIT – II
International Trade Policy

Theory of Tariffs: Partial Equilibrium analysis of Tariff (both small country and large country case), General Equilibrium analysis of a Tariff (both small country and large country case). Optimum tariff.

Non-Tariff Barriers and Neo-protectionism.

Economic Integration: Theory of Customs Unions. Static effects (Trade creation and trade diversion). Dynamic effects of custom unions.

International Trade Agreements: Multilateralism: The World Trade Organization (WTO), Regional Trade Agreements (RTAs), Trade, Labour, and Environmental Standards

UNIT – III
The Balance of Payments

Concept and Components of Balance of Payment.

The Price Adjustment Mechanism with Flexible and Fixed Exchange Rates, Marshall-Learner conditions, J-curve effect, Gold Standard (Price-Species Flow Mechanism).

The Income Adjustment Mechanism, Foreign Trade Multiplier.

Open-Economy Macroeconomics and Adjustment Policies: Equilibrium in the Goods Market, in the Money Market and in the Balance of Payments (Mundell-Fleming Model).

UNIT – IV
Foreign Exchange Markets and International Monetary System:

Foreign Exchange Rates, Arbitrage, Spot and Forward Rates, Currency Swaps, Futures and Options, Foreign Exchange Risks, Hedging and Speculation. Euro currency Markets.

The International Monetary System: Past, Present and Future.

IMF and World Bank: Evolution and current relevance

Recommended Readings:

Bhagwati, J. (1987). International trade: Selected readings. (Vol. 1). MIT Press.

Clement, M.O., Pfister, R. L., & Rothwell, K. J. (1967). Theoretical issues in international economics. Constolete and Company Ltd.

Dunn, R.M., & Mutti, J. H. (Latest Edition). International economics. New Jersey: John Wiley & Sons. Inc.

Jones, R. W., & Kenen, P.B. (1984). Handbook of international economics. Vol. I, II and III. New York: Elsevier.

Krugman, P.R., & Obstfeld, M.(Latest Edition). International economics: Theory and policy. United States: Pearson Publication.

Salvatore, D. (Latest Edition). International economics. New Jersey: John Wiley& Sons Inc. Souderton, B., & Geoffrey R. (1994). International economics. London : McMillan.

Further Readings:

Ellsworth, P T & Clark L. (1975). The International economy. London: McMillan Press.

Mikic, M. (1998). International trade. London: McMillan Press.

Peter H L. (1969). International economics. New Jersey : Princeton University Press. Shenkar, O., Luo, Y., & Chi, T. (2014). International business. London: Routledge.

MAECODSE-105 (A) - FUNDAMENTAL ISSUES IN POLITICAL ECONOMY

Topology of the code	**Discipline Specific Elective**
Level of the Course	**400**
Course Duration /LTP (Lectures, Tutorials,Practical)	**L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment 20 marks**
Time	**3hours**

Course Objective:-The course aims to provide a rigorous and critical understanding of political economy, tracing its evolution from classical capitalism to contemporary globalization. It seeks to familiarize students with Marxian critiques of capitalism, the economic role of the state, and major theories of regulation.

PedagogyoftheCourseOutcomes: The course will be delivered primarily through classroom-based teaching (80%), consisting of lectures, guided discussions, and conceptual exposition of core theoreticalframeworks.The remaining 20% will be devoted to internal assessment, which may include written assignments, presentations, and critical reviews aimed at reinforcing analytical skills and independent engagement with political economy literature.

CourseOutcomes: Upon completion of this course, students will develop a critical understanding of major debates in political economy, enabling them to compare different economic systems and theoretical traditions. They will be able to explain and apply Marxian analysis to examine the functioning and contradictions of capitalism, assess the role of the state through regulatory and institutional frameworks, and critically evaluate the impacts of globalization and neoliberal policies on agrarian relations.

INSTRUCTIONSFORPAPER-SETTERANDCANDIDATES:

The maximum marks for the paper will be 100. The question paper will be of 80 marks and continuous evaluation 20 marks. Time allowed will be 3 hours.

The paper-setter must put a note in the question paper in this regard.

1. Thereshallbe**9**questionsinall.

2. The first question is **compulsory** comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10 x 2= 20). Rest of the paper shall contain four units.

3. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60)

Unit I

Foundations of Political Economy

Scope and subject matter of political economy; Mode of production and the superstructure; Pre-capitalist modes of production: Primitive communism, Slavery and Feudalism

Unit II

Capitalist Mode of Production

Core features of the capitalist mode of production**;** Capital accumulation and the reserve army of labour**;** The falling tendency of the rate of profit**;** The nature and dynamics of capitalist crises

Unit III

State, Monopoly Capitalism, and Accumulation

The state and its role in capital accumulation under capitalism; Development of monopoly capitalism: Concentration and centralisation of capital, Corporations, cartels, trusts, and mergers, Banks and financial institutions in monopoly capitalism

Unit IV

Agrarian Questions and Imperialism in 21st Century

The agrarian question in classical and contemporary debates; Capitalism in agriculture

Imperialism: Classical theories (Hobson, Lenin, Luxemburg); Contemporary imperialism;

Contemporary imperialism and its relationship to the agrarian question

Recommended Readings

Unit I

Lange, O. (1963). *Political Economy*, Vol. I. Chapters 1–2.

Nikitin, P. I. (1983). *The Fundamentals of Political Economy*. Chapter 1.

Unit II

Sweezy, P. M. (1942). *The Theory of Capitalist Development*. Chapters 4, 5, 6, and 8.

Foley, D. K. (1986). Understanding Capital. Cambridge, MA: Harvard University Press. Chpater 4.

Upadhyay, V., & Singh, P. (Eds.). (2021). *Global Political Economy: A Critique of Contemporary Capitalism*. Routledge. Chapters 2 and 4.

Howard, M. E., Howard, M. C., & King, J. E. (1988). *The Political Economy of Marx*. NYU Press. Part IV

Unit III

Sweezy, P. M. (1942). *The Theory of Capitalist Development*. Chapter 13.

Baran, P. A., & Sweezy, P. M. (1966). *Monopoly Capital*. NYU Press. Introduction and Chapter 8.

Upadhyay, V., & Singh, P. (Eds.). (2021). *Global Political Economy: A Critique of Contemporary Capitalism*. Routledge. Chapters 6 and 11.

Fine, B. and Saad-Filho, A. (2016) *Marx's Capital* . London: Pluto Press. Chapter 12.

Unit IV

Kautsky, K. (1988). *The Agrarian Question*, Vol. I. Introduction to the English Edition by Hamza Alavi & Teodor Shanin and Part I.

Patnaik, U. (Ed.). (2007). *The Agrarian Question in Marx and His Successors, Vol. I: The Development of Capitalist Relations in Agriculture*.

Hunt, E. K., & Lautzenheiser, M. (2011). *History of Economic Thought*. Chapter 13.

Amin, S. (2012). "Contemporary Imperialism and the Agrarian Question." *Agrarian South: Journal of Political Economy*, 1(1), 11–26.

Additional Readings

Bernstein, H. (2016). Agrarian questions then and now. In *Agrarian Questions* (pp. 22-59). Routledge.

Dobb, M. (1975). *Theories of Value and Distribution Since Adam Smith*.

Harvey, D. (2010). *A Companion to Marx's 'Capital'*. London: Verso

Heilbroner, R. L. (2011). *The Worldly Philosophers: The Lives, Times, and Ideas of the Great Economic Thinkers*. Simon & Schuster.

Hiferding, R. (2019). *Finance capital: A study in the latest phase of capitalist development*. Routledge.

Lenin, V. I. (1917). *Imperialism, the Highest Stage of Capitalism*.

Marx, K. (1992). *Capital*, Vol. I. Penguin Books.

Patnaik, P. (2014). "Imperialism and the Agrarian Question." *Agrarian South: Journal of Political Economy*, 3(1), 1–15.

Shaikh, A. (2016). *Capitalism: Competition, Conflict, Crises*. Oxford University Press

MAECODSE-105 (B): LABOUR ECONOMICS

Topology of the code	**Discipline Specific Elective**
Level of the Course	**400**
Course Duration /LTP (Lectures, Tutorials,Practical)	**L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment 20 marks**
Time	**3hours**

Objective: This course aims to provide a comprehensive understanding of labour market theories, from classical to contemporary perspectives. It examines the functioning of labour institutions, wage determination, and the impact of legislation and industrial relations. The course also evaluates contemporary challenges in the Indian labour market, including informalisation, gender disparities, and the evolving nature of work.

Outcomes: This course shall enable students to comprehend the theoretical and empirical foundations of labour markets and employment dynamics. It shall provide insights into the

functioning of wage determination mechanisms, labour market segmentation, and institutional frameworks governing labour relations. The course shall also equip students to critically analyse contemporary issues in the Indian Labour Market such as informalisation, wage inequality, gender disparities, and structural transformations, thereby preparing them for advanced research and policy analysis in labour and employment economics.

Pedagogy of the Course Work: The students would be evaluated at the end of each semester through subjective-type questions/answers (both short and essay type). The scripts would be evaluated by the teacher/instructor teaching the paper and the grade shall be submitted accordingly. The course relies on a combination of lectures, solving problems and discussing academic articles or real-life situations. The teacher will assign topics for assignments on contemporary themes and issues from the syllabi.

Instructions For the Paper-Setter and Candidates:

1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question, which would be compulsory, shall be a short-answer type (word limit 2530 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short-answer type questions. Each short-answer-type question would carry 2 marks (10× 2 = 20).
4. The rest of the paper shall contain 4 units. Each unit shall have two questions, and the candidates shall be required to attempt one question from each Unit. Each question shall carry 15 marks (15 × 4 = 60).

UNIT-I

Labour Markets and Employment: Nature and characteristics of labour markets in developing countries

like India; Paradigms of labour market analysis – Classical, neo-classical and dualistic economy; Demandfor labour; Supply of labour in relation to growth of labour force; Labour market policies.

Employment and Development Relationship: Poverty and unemployment in developing countries; Unemployment – Concept, Types, and Measurement, particularly in India; Impact of rationalization, technological change and modernization on employment.

UNIT-II

Wage Determination: Classical, Neo-classical and bargaining theories of wage determination; Concepts of minimum wage, living wage and fair wage in theory and practice; Discrimination in labour markets; Wage determination in various sectors – rural, urban, organized, unorganized and informal sectors; Non-wage component of labour remuneration; Inflation-wage relationship at micro and macro levels; Productivity and wage relationship; Analysis of rigidity in labour markets; Asymmetric information and efficiency of labour markets in wage determination; National wage policy.

UNIT-III

Industrial Relations: Theories of Labour Movement – Growth, pattern and structure of labour unions in India, Achievements of labour unions; Causes of industrial disputes, their settlement and prevention mechanism; Role of tripartism; Current trends in collective bargaining.

UNIT-IV

State and Labour: State and Social Security of Labour – Concept of Social Security and its evolution; Social assistance and social insurance; Review and appraisal of State policies with respect to social security and labour welfare in India; Special problems of labour; Receding State and its effect on working of labour markets; Labour market reforms – Exit policy, need for safety nets, measures imparting flexibility in labour markets; Second National Commission on Labour; Globalization and Labour Markets.

Recommended Readings:

Binswanger, H. P. & M. R. Rosenzweig, Contractual Arrangements, Employment, and Wages (eds.) (1984) in Rural Labour Markets in Asia, Yale University Press, New Haven.

Datt, G. (1996), Bargaining Power, Wages and Employment: An Analysis of Agricultural Labour Markets in India, Sage Publications, New Delhi.

Deshpande, L. K., P. R. Brahmananda, Employment Policy in a Developing Economy, and EAG Robinson (eds.)(1983) Vols. I & II, Macmillan, London.

Deshpande, L. K. and J. C. Sandesara, Wage Policy and Wages Determination in (eds.) (1970) India, Bombay University Press, Bombay.

Deshpande, S. G. Standing and L. K., Labour Flexibility in a Third World Metropolis, Deshpande (1998) Commonwealth Publishers, New Delhi.

Foster, A. D. & M. R. Rosenberg (1993), Information Flows and Discrimination in Labour Markets in Rural Areas in Developing Countries, Annual Conference on Development Economics, World Bank, Washington D C.

Hauseman, S. (1991), Industrial Restructuring with Job Security, Harvard University Press, Cambridge, Mass.

Hicks, J. R. (1932), The Theory of Wages, Clarendon Press, Oxford.

Lester, R. A. (1964), Economics of Labour (2nd edition), Macmillan, New York.

McConnell, C. R. & S. L. Brue (1986), Contemporary Labour Economics, McGraw-Hill, New York.

McCormick, B. & Smith (eds.) (1968), The Labour Market, Penguin, Harmondsworth.

MAECODSE-105 (C) – OPTIMISATION TECHNIQUES AND GAME THEORY

Topologyofthecode	**Discipline Specific Elective**
LeveloftheCourse	**400**
Course Duration /LTP (Lectures,Tutorials,Practical)	**L=45hours (3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment 20 marks**
Time	**3hours**

Objective: The paper aims to make the students familiar with some of the mathematical techniques which have been developed in the last few decades to solve the basic and one of the fundamental problems faced by economic agents, i.e. economizing at a particular point of time or over a period of time.

Pedagogy: The course will be delivered through a combination of classroom teaching (80%) and computer-based practical training with viva-voce examination (20%). Classroom instruction will focus on conceptual clarity, mathematical derivations, and economic interpretation, while computer practical sessions will emphasize hands-on application of optimization and game-theoretic techniques using MATLAB, reinforcing analytical and computational skills.

Course Outcome:
Upon successful completion of the course, students will be able to apply static and dynamic optimization techniques to economic problems, analyze constrained decision-making using nonlinear and linear programming methods, and interpret economic applications of calculus of variations and optimal control theory. They will develop the ability to model strategic interactions using game-theoretic concepts, identify equilibrium outcomes in both static and dynamic settings, and implement optimization and game-theoretic solutions computationally using MATLAB for applied economic analysis.

Instructions for Paper-setter and candidates:
• The maximum marks for the paper will be 100.
• The theory question paper will be of 80 marks and computer practical of 20 marks.
• Time allowed in theory paper will be 3 hours.
• The computer practical will be conducted by the teacher concerned in the computer labof the department and marks in practical paper will be awarded in followingcategories:
a) 10 marks to successful execution of the program/assignment on computer usingMatLab. Each student will be given 1 hour to run the given computer program inMatLab;
b) 5 marks to File work on MatLab programming; and
c) 5 marks to viva-voce on practical application of OTGT.

The paper-setter must put a note in the question paper in this regard.

1. Thereshallbe**9**questionsinall.

2. The first question is **compulsory** comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10 x 2= 20). Rest of the paper shall contain four units.

3. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60)

UNIT-I

Topics in optimization: Nonlinear programming and Kuhn-Tucker condition. Theconstraint qualification. Economic Applications: Peak Load pricing and war time rationing.

Topics in linear Programming: Dual Simplex Method, Sensitivity and ParametricAnalysis.

UNIT-II

Dynamic Optimization: A Brief introduction to the calculus of variations (Eulerequation),end point (transversality) conditions. The control problem, Statement and types of control.

Pontryagin's Maximum principle, finite and infinite time horizon, interpretation ofhamiltomian, costate variables, equation of motion; end point conditions. Constructing aphase diagram and its analysis. Economic Applications of Calculus of variations andOptimum Control theory.

UNIT-III

Game Theory*:* Two persons Zero Sum Game, Max-Min Strategy. Non-Zero Sum Game,Static Games in Complete Information, Strongly and Weakly Dominant Strategies, DominantStrategy Equilibrium; Pure and Mixed Strategies;

Concept of Nash Equilibrium-Examplessuch as Prisoner's Dilemma, Battle of Sex, Matching Pennies, Co-ordination Problem andMultiple Equilibria, etc., Focal Points, Dynamic Games in Complete Information, Sub-Gamesand Sub-Game Perfect Nash Equilibrium.

UNIT-IV

Optimization using MATLAB*:* Basic Commands of MATLAB, Solving Matrices using MATLAB, Solving Linear Programming, Non-Linear Programming and Dynamic Optimization problems Using MATLAB. Solution of Games in MATLAB through Linear Programming.

Essential Readings

Chiang, A. C. (1984). *Fundamental methods of mathematical economics*. McGraw-Hill.

Chiang, A. C. (1999). *Elements of dynamic optimization*. Waveland Press.

Dixit, A. (1976). *Optimization in economic theory*. Oxford University Press.

Gibbons, R. (1992). *A primer in game theory*. Pearson Education.

Gilat, A. (2007). *MATLAB: An introduction with applications*. Wiley India.

Gravelle, H., & Rees, R. (1992). *Microeconomics*. Pearson Education.

Kirani, S. Y. (2007). *MATLAB programming*. Prentice Hall of India.

Osborne, M. J. (2004). *An introduction to game theory*. Oxford University Press.

Sundaram, R. K. (1996). *A first course in optimization theory*. Cambridge University Press.

Yang, W. Y., Cao, W., Chung, T.-S., & Morris, J. (2007). *Applied numerical methods using MATLAB*. Wiley India.

Further Readings

Dixit, A., & Skeath, S. (2004). *Games of strategy* (2nd ed.). W. W. Norton & Company.

Intriligator, M. D. (1971). *Mathematical optimization and economic theory*. Prentice Hall.

Koopmans, T. C. (1957). *Three essays on the state of economic science*. McGraw-Hill.

Lancaster, K. (1965). *Mathematical economics*. Macmillan.

Osborne, M. J., & Rubinstein, A. (1994). *A course in game theory*. MIT Press.

Rasmusen, E. (2000). *Games and information: An introduction to game theory* (3rd ed.). Blackwell Publishers.

SEMESTER – II

M.A. Economics with Research			
Level of Course	**Course Code**	**Course Name**	**Credits**
L-400	MAECODSC-201	Advanced Microeconomics-II	4
	MAECODSC-202	Advanced Macroeconomics-II	4
	MAECODSC-203	Advanced Quantitative Methods for Economics-II	4
	MAECODSC-204	Advanced Public Finance	4
	MAECODSE-205 (A) or MAECODSE-205 (B) or MAECODSE-205 (C)	Advance Issues in Political Economy or Environmental Economics or Economics of Money and Banking	4

MAECODSC-201: ADVANCED MICROECONOMICS-II

Topology of the code	**Discipline Specific Core**
Level of the Course	**400**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L=60 hours(4 hours aweekfor15weeks=01 semester), T=00 hours, P=0**
Credits	**4**
Max. Marks	**100Marks=Theory80marks+InternalAssessment20 marks**
Time	**3hours**

Objectives:

Theobjectiveofthispaperistorigorouslyandcomprehensivelyequipthestudentswiththeoreticalconcepts, methodology and process of reasoning involved in analyzing economic behavior of individuals, firms and markets using, ingeneral, astaticandpartial equilibriumframework.

The students would be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Pedagogy of the Course Work:

The time–tested method of imparting verbal instructions through lectures would be used. Examples, in sofar as possible, would be selected from everyday life/experience. The feedback would be sought throughthe students' written assignments, tutorials and seminars.The students would be evaluated at the end of the academic year through subjective type questions/answers

(both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Course outcomes:
Upon completion of this course, students will be able to analyse firm and industry behaviour under different market structures, including perfect competition, monopoly, monopolistic competition, and oligopoly. They will understand pricing strategies, output determination, welfare implications, and the role of public policy in regulating markets. Students will be able to apply game theory to strategic decision-making and cooperation among firms. The course will also enable students to evaluate factor pricing mechanisms and use welfare economics to assess efficiency, equity, and policy responses to market failures such as monopoly power and externalities.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:
1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question would be compulsory, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks (10x2=20).
4. Rest of the paper shall contain 4 units. Each Unit shall have two questions and the candidates shall be required to attempt on question from each Unit – 4 in all. Each question shall carry 15 marks (15 x 4 = 60).
5. Candidates are allowed to use Simple, Non-Scientific and Non-Programmable Calculator in the Examination Hall.
6. It is compulsory for the paper setter to mention the requirement of log, antilog and statistical tables.

UNIT-I

Analysis of Competitive Markets: Profit Maximization and Equilibrium of a Firm and Industry in the Short Run and Long Run under Perfect Competition including the Supply Curve. Consumers' and Producers' Surplus and Efficiency of Perfect Competition. Welfare Effects of an Excise Tax, Import Tariff, Price controls (supports) and Production Quotas.

Monopoly: Source, Measurement and Welfare Costs and Dynamic benefits of Monopoly Power. Monopoly Price Discrimination. Output and Welfare Comparisons with Single Price Monopoly. Inter–temporal Price Discrimination and Peak Load Pricing. Two part tariffs and Tie-in Sales. Public Policy toward Monopoly

UNIT-II

Monopolistic Competition: Characteristics. A firm's Short and Long Run Equilibrium with Differentiated Products and Selling Costs (Advertisement). Group equilibrium. The Issue of Excess Capacity under Monopolistic Competition. The debate over Advertising.

Oligopoly: Cournot, Stackellberg and Kinked demand Curve. Leadership Models. Collusion and Cartels.
Game Theory and Economics of Cooperation: Dominant Strategy and Nash Equilibrium. The Prisoner's Dilemma, Mixed Strategies, Repeated Games, Collusion and Tit-for-Tat Strategy. Public Policy towards Oligopoly. Contestable Markets.

UNIT-III

Markets for Factor Inputs: Factor Pricing under Perfect Competition: Demand for a Factor Input when One Input is Variable. Demand for a Factor Input when Several Inputs are Variable. Market Demand for a Labour. Supply of Labour by an Individual: Trade Off between Work and Leisure. Wage rate determination.

Factor Pricing under Imperfect Competition: Wage rate determination with Monopolistic Power in Product Market, Monoposonistic Power in Factor Market; Bilateral Monopoly. Competitive Buyer Firm versus Monopoly Union.

UNIT-IV

Welfare Economics: Social Welfare Function: Utilitarian, Rawlsian and Bergson-Samuelson Social Welfare Functions and Optimization of Social Welfare.

General Equilibrium and Efficiency: Pareto-Optimality Conditions. Perfect Competition and Economic Efficiency. First and Second Theorem of Welfare Economics. Compensation Criteria and Arrow's Impossibility Theorem.

Externalities & Efficiency: Externalities in Production and Consumption. Coase Theorem and Public Policies towards Externalities. Common Property: Overuse of Common Property, Solving Common Problems

Essential Readings:

Boumol, W. J. & Blinder, A. S. (2007). *Macroeconomics: Principles and Policy*, (9thed.). Thomson.
Ferguson, C. E. and Gold J. P. (1989). *Micro Economic Theory* (6th ed.). All India Book Sellers.
Mankiw, G. N. (2017). *Principles of Economics,* (3rd ed.). Thomson.
Pindyck, Robert, S., Rubinfel, D. L. & Gupta, P.L. (2017). *Micro-Economics*. New Delhi: Prentice Hall of India.
Salvatore, D. (2007). *Microeconomics: Theory and Applications* (4th ed.). Oxford University Press.

Further Readings:

Briet, W. & Hochman, H.M. (1977). *Readings in Micro Economics.* Hindsala: Dryden Press.
Koutsoyiannis, A. (2016). *Modern Micro Economics.* London: McMillan Press.
Maddala, G. S. & Miler, E. (2004). *Micro-Economic Theory and Applications.* Tata McGraw Hill.
Ryan, W. J. L. & Pearce, D. W. (1989). *Price Theory.* New Delhi : McMillan India.

MAECODSC-202: ADVANCEDMACROECONOMICS-II

Topology of the code	**Discipline Specific Core**
Level of the Course	**400**
Course Duration/LTP (Lectures, Tutorials, Practical)	**L=60 hours(4 hours aweekfor15weeks=01 semester), T=00 hours, P=0**
Credits	**4**
Max. Marks	**100Marks=Theory80marks+InternalAssessment20 marks**
Time	**3hours**

Objectives:

Macroeconomicsoraggregativeeconomicanalysisestablishesthefunctionalrelationshipbetween the large aggregates. The aggregate analysis has assumed such a great significance in recent times that a prior understanding of macroeconomic theoretical structure is considered essential for the proper comprehension of the different issues and policies. Macroeconomics now is not only a scientificmethod of analysis; but alsoa bodyof empirical economicknowledge.

Thepaperentitled"Macro-Economics-I"equipsthestudentsatthepostgraduateleveltounderstand systemic facts and latest theoretical developments for empirical analysis.

The students would be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Pedagogy of the Course Work:

The time–tested method of imparting verbal instructions through lectures would be used. Examples, in sofar as possible, would be selected from everyday life/experience. The feedback would be sought throughthe students' written assignments, tutorials and seminars.The students would be evaluated at the end of the academic year through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Course outcomes:

Upon completion of this course, students will be able to analyze macroeconomic equilibrium in closed and open economies using the IS–LM–BP framework and assess the effectiveness of monetary and fiscal policies under different exchange rate regimes. They will understand the causes, effects, and theories of inflation and evaluate the Phillips Curve debate between Keynesian and Monetarist perspectives. Students will be able to explain major trade cycle theories and models of economic fluctuations. The course will also enable students to critically assess New Classical and New Keynesian approaches, including rational expectations, business cycle theories, and wage–price rigidities, in the formulation of macroeconomic policy.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:

1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question would be compulsory, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to

attempt any 10 short answer type questions. Each short answer type question would carry 2 marks (10x2=20).
4. Rest of the paper shall contain 4 units. Each Unit shall have two questions and the candidates shall be required to attempt on question from each Unit – 4 in all. Each question shall carry 15 marks (15 x 4 = 60).
5. Candidates are allowed to use Simple, Non-Scientific and Non-Programmable Calculator in the Examination Hall.
6. It is compulsory for the paper setter to mention the requirement of log, antilog and statistical tables.

UNIT-I

IS and LM Framework: Derivation, Properties, Shifts and Rotations of IS and LM Curves under closed and open economy systems. Derivation, Properties, Shifts, and Rotations of BP Curve. Simultaneous Equilibrium in Money and Product Markets. Impact of Opening-up on simultaneous equilibrium (i.e., IS-LM-BP simultaneous equilibrium).

Monetary and Fiscal Policies: Objectives, Conflicts among Objectives. Relative Effectiveness of Monetary and Fiscal Policies under Different Situations in IS-LM-BP Framework. International Trinity and Quadrilemma choices under IS-LM-BP framework.

UNIT-II

Inflation: Effects of Inflation. Theories of Inflation: Quantity Theory, Keynesian Theory, Monetarist views on Inflation, Modern theory of Inflation, Structural Theory. The menu of policy choices.
Philips Curve Analysis: Short Run and Long Run views. The Monetarist-Keynesian Debate and the Phillips Curve.

UNIT-III

Trade Cycle: Hansen–Samuelson Accelerator–Multiplier Interaction Model, Hicks Model, Kaldor Model, Goodwin model of endogenous cycles.

UNIT – IV

The New Classical School: Rational Expectations Hypothesis: Dynamic Time Inconsistency, Policy Ineffectiveness Proposition.

The Random Walk of GDP: The Relative Importance of AD and AS.

Real Business Cycle Model: Disturbances and Propagation mechanism. Macroeconomic Policy in Real Business Cycle Model.

The New-Keynesian School: Real and Nominal Wage-Price Rigidity Models - Menu Costs Model, Implicit Wage Contract Models, Efficiency Wage Models, Insider-Outsider Models.

Suggested Readings:

Ackley, G. (2007). Macroeconomic Theory. Macmillan.
Allen, R.G.D. (1967). Macro-Economic Theory: A Mathematical Treatment. Palgrave Macmillan.
Blanchard, O. (2009). Macroeconomics. New Delhi: Pearson Education Inc.
Blanchard, O.J. & Fischer. (1996). Lectures on Macroeconomics, Delhi: PHI Learning Pvt. Ltd.
Branson, W. H. (1989). Macro Economic Theory and Policy (3rd ed.). Harper Collins.
D'Souza, E. (2008). Macroeconomics. Pearson Education.
Dornbusch, R., Fisher, S & Startz, R. (2002). Macroeconomics, (9th ed.). New Delhi: Tata McGraw Hills.
Levacic, R. & Rebmann, A. (1982). Macroeconomics: An Introduction to Keynesian Neoclassical Controversies. Macmillan.
Miller, R. L.& Pulsinelli, R. (1986). Macroeconomics. New York: Harper & Row.
Ola Olsson (2010). Essentials of Advanced Macroeconomic Theory, Routledge.
Romer, D. (1996). Advanced Macroeconomics. McGraw-Hill.
Abel, A. B., Bernanke, B. S. (2007). Macroeconomics. New Delhi: Pearson Education Inc.
Dornbusch, R. (1980). Open Economy Macroeconomics. New York: Weltwirtschaftliches Archive.
Frisch, H. (1983). Theories of Inflation. Cambridge: Cambridge University Press.
Froyen, R. T. (2002). Macro Economics. New Delhi: Pearson Education Inc.
Gordon, R. J. (2012). Macroeconomics. Delhi: PHI Learning Pvt. Ltd.
Heijdra, B. J. (2017). The Foundations of Modern Macroeconomics. Oxford University Press.
Hicks, J. R. (1950). A Contribution to the Theory of Trade Cycle Oxford: Clarendon Press.
Lucas, R. (1981). Studies in Business Cycle Theory. , Massachusetts: MIT Press Cambridge.
Makin, A. J. (2002). International Macroeconomics. Princeton Hall.
Mankiw, N. G. & Romer, D. (1991). New Keynesian Economics. MIT, 2 Volumes.
Mathews, R. C. O. (1964). The Trade Cycle. Cambridge: Cambridge University Press.
Pentecost, E. (2000). Macroeconomics – An Open Economy Approach. MacMillan Press Ltd.
Raghbendra, J. (1964). Contemporary Macroeconomics: Theory and Policy. New Age International ublishers.
Sherffrin, S. (1999). Rational Expectations. Cambridge University Press.
Snowden, B & Vane, H. R. (2005). Modern Macroeconomics: Its Origin, Development and Current Stat., UK: Edward Press.
Wendy, Carlin/Soskice, D. (2007). Macroeconomics: Imperfections, Institutions and Policies. Oxford University Press.

MAECODSC-203: ADVANCEDQUANTITATIVE METHODS FOR ECONOMICS – II

Topology of the code	**Discipline Specific Core**
Level of the Course	**400**
Course Duration/LTP (Lectures, Tutorials, Practical)	**L=60 hours(4 hours aweekfor15weeks=01 semester), T=00 hours, P=0**
Credits	**4**
Max. Marks	**100Marks=Theory80marks+InternalAssessment20 marks**
Time	**3hours**

Objectives:

The objective of this paper is to train the students in the use of mathematical and statistical tools in analysing economic problems. The course content contains simple tools and techniques, thought necessary for data collection, presentation, analysis and drawing inferences about various statistical hypotheses.

Pedagogy of the Course Work:

The time–tested method of imparting verbal instructions through lectures would be used. Examples, in sofar as possible, would be selected from everyday life/experience. The feedback would be sought throughthe students' written assignments, tutorials and seminars.The students would be evaluated at the end of the academic year through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Course outcomes:

Upon successful completion of this course, students will be able to apply techniques of calculus, differential and difference equations, and linear programming to analyse and solve economic problems. They will develop proficiency in probability distributions, statistical inference, and hypothesis testing for informed decision-making under uncertainty. Students will be able to formulate and interpret economic models related to demand, cost, revenue, market equilibrium, and optimization. The course also enables learners to analyse real-world data using estimation, testing, and variance analysis techniques. Overall, students will acquire strong quantitative and analytical skills essential for economic research and policy analysis.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:

1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question would be compulsory, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks (10x2=20).
4. Rest of the paper shall contain 4 units. Each Unit shall have two questions and the candidates shall be required to attempt on question from each Unit – 4 in all. Each question shall carry 15 marks (15 x 4 = 60).

5. Candidates are allowed to use Simple, Non-Scientific and Non-Programmable Calculator in the Examination Hall.
6. It is compulsory for the paper setter to mention the requirement of log, antilog and statistical tables.

UNIT - I

Integration: Indefinite and Definite, Integration of a Function of only one Variable; Methods of Integration; Substitution and Partial Fraction. Application of Integration in Economics; Consumers' and Producers' Surplus; Finding Demand Function, Maximum Profit and Capital Growth Equation.
Differential Equations: Introduction, Solution of Differential equations – Variable-separable, Homogenous, Exact, Linear and Bernoulli's form cases; Solution of Linear Differential Equation with Constant Coefficients.
Economic Applications w.r.t. Demand, Revenue, Cost Functions and Simple Market Equilibrium Models.

UNIT – II

Difference Equations: Solution of a Difference Equation and Simple Economic Applications.

Linear Programming: Definition, Assumptions, Formulation and Solution of LPP (Feasible, Basic and Optimal Solutions) by graphical and simplex method. Dual Problem: Solution of Primal and Dual Problems by Simplex Method and their Interpretation.

UNIT – III

Probability Distributions: Properties (including derivation of mean and variance) and Applications of Binominal, Poisson and Normal Distributions.

Statistical Inferences: Concepts of parameter, statistic and standard error (including its utility). Sampling Distribution of a Statistic, Distribution of Sample Mean, Properties of an Ideal Estimator; Unbiasedness, Consistency, Efficiency and Minimum Variance. Interval Estimation and Confidence Interval for mean, difference of means, single proportion and difference of proportions.

UNIT – IV

Tests of Significance: Large and Small Sample Tests based on Normal and t Distribution, Test of Significance of Mean, Difference between means, Single Proportion, Difference between two proportions, Simple Correlation Coefficient and Simple Regression Coefficient, test based on F and Z Distribution (Fisher's Z Statistics).

Chi-Square Tests: Conditions for Chi-Square Test; Application of χ^2 Distribution – Test for Goodness of Fit, Independence and Specified Value of the Variance.

Analysis of Variance: Applications with regard to one way and two way classifications.

Suggested Readings:

Allen, R. G. D. (latest edition). *Mathematica*l *Analysis for Economists*. New Delhi: Trinity Press.

Allen, R. G. D. (1986). *Mathematica*l *Analysis for Economists.* Macmillan Publisher India Ltd.

Arora, P. N., Arora, S., Arora, S. & Arora, A. (2007). *Comprehensive Statistical Methods.* New Delhi: S. Chand & Company Ltd.

Chiang, A. C. &Wainwright, K. (2005). *Fundamental Methods of Mathematical Economics* (4th ed.). New York: McGraw Hill.

Devore, J. L. (2012). *Probability and Statistics for Engineering and the Sciences*, (8th ed.), Cengage Learning India.

Freund, J. E. (1992). *Mathematical Statistics*. (5th ed.). New Delhi: Prentice-Hall, Cengage Learning, India.

Gupta, S. C. (latest edition). *Fundamentals of Statistics* (7th ed.). Himalaya Publications.

Gupta, S. P. (2014). *Statistical Methods.* New Delhi: Sultan Chand & Sons.

Kandoi, B. (2012). *Mathematics for Business and Economics with Applications*, Vol. 1. Mumbai: Himalaya.

Simon, C. P. & Lawrence, B. (2006). *Mathematics for Economists* (Indian ed.). Viva –Norton Student edition: Norton and Company.

Spiegel, M. R. & Stephens L. J. (2000). *Theory and Problems of Statistics:* Schaum's Series, *(*3rd ed.).

New Delhi: Tata McGraw Hill.

Weber, J. E. (1982). *Mathematical Analysis (Business and Economic Applications)* (4th ed.).

New York: Harper and Row.

Yamane, T. (2006). *Mathematics for Economist-an Elementary survey* (2nd ed.). New Delhi: Prentice Hall.

Yamane, T. (1972). *Fundamental Methods for Mathematical Economics*. New Delhi: Prentice-Hall.

MAECODSC-204: ADVANCED PUBLIC FINANCE

Topology of the code	**Discipline Specific Core**
Level of the Course	**400**
Course Duration/LTP (Lectures, Tutorials, Practical)	**L=60 hours(4 hours aweekfor15weeks=01 semester), T=00 hours, P=0**
Credits	**4**
Max. Marks	**100Marks=Theory80marks+InternalAssessment20 marks**
Time	**3hours**

Objectives:

Role and functions of the Government in an economy have been changing with the passage of time.The term 'Public Finance' has traditionally been applied to the package of those policies andoperations which involve the use of tax and expenditure measures while budgetary policy is animportant part to understand the basic problems of use of resources, distribution of income etc. Thereare vast array of fiscal institutions – tax systems, expenditure programmes, budgetary procedures,stabilization instruments, debt issues, levels of government etc., which raise a spectrum of issuesarising from the operation of these institutions. Further, the existing of externalities, concern foradjustment in the distribution of income and wealth, etc. require political processes for their solutionin a manner which combines individual freedom and justice. This paper combines a thoroughunderstanding of fiscal institutions with a careful analysis of the issues which under line budgetarypolicies in general and Indian experience in particular.The students would be evaluated at the end of each semester through subjective typequestions/answers (both short and essay type). The scripts would be evaluated by the examinershaving adequate postgraduate teaching experience in the paper/options concerned.

Pedagogy of the Course Work:

The time–tested method of imparting verbal instructions through lectures would be used. Examples, in sofar as possible, would be selected from everyday life/experience. The feedback would be sought throughthe students' written assignments, tutorials and seminars.The students would be evaluated at the end of the academic year through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Course outcomes:

After completing this course, students will be able to explain the scope and theoretical foundations of public finance, analyse equity and efficiency issues in taxation and public expenditure, assess the macroeconomic implications of deficit financing and public debt, and critically evaluate India's fiscal institutions, intergovernmental financial relations, and rule-based fiscal management in light of contemporary economic challenges.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:

1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question would be compulsory, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to

attempt any 10 short answer type questions. Each short answer type question would carry 2 marks (10x2=20).
4. Rest of the paper shall contain 4 units. Each Unit shall have two questions and the candidates shall be required to attempt on question from each Unit – 4 in all. Each question shall carry 15 marks (15 x 4 = 60).
5. Candidates are allowed to use Simple, Non-Scientific and Non-Programmable Calculator in the Examination Hall.
6. It is compulsory for the paper setter to mention the requirement of log, antilog and statistical tables.

UNIT – I

Scope of Public Finance, Provision of Private Goods and Public Goods in Partial and General Equilibrium (Pareto's Optimality criteria.)

Equity in Distribution. Various approaches to distributive Justice. Public Choice and Fiscal Politics. Alternate Voting rules.

UNIT – II

Various Approaches of Equity in Taxation: Benefit Principle including Lindahl Theory. Ability to Pay Approach.

Incidence analysis of various taxes in product and factor markets.

Effects of Taxation on Work Effort, Savings and Investment.

UNIT - III

Deficit Financing: Concept and its relation with Inflation, Deficit Financing in India.

Issues relating to Public Debt: Debt Burden Analysis and Management of Public Debt, Domar's concept of Debt Sustainability Public Debt in India.

Need for rule based fiscal consolidation. Fiscal Responsibility and Budget management (FRBM) act, 2003. Recent amendments to FRBM act.

UNIT – IV

Theories of Public Expenditure: Wagner's Law and Peacock - Wiseman Hypothesis. Structure and Classification of Public expenditure in India.

Principles of Multiunit Finance. Financial aspects of 73rd and 74th Constitutional Amendments.

Centre – State Financial Relations in India: Assessment of Horizontal and vertical imbalances. Constitutional provisions relating to fiscal and financial powers of the States, FinanceCommissions and their formulae for sharing taxes, Financial aspect of Sarkaria Commission Report.

Essential Readings:

Atkinson, A.B., & Stiglitz, J.E.(2007). *Lectures on public economics*. New Jersey: Princeton University Press.

Bhatia, H. L. (2006). *Public finance*. Vikas Publishing House Private Ltd.

Cullis, J., Jones, P., & Jones, P.R. (2009). *Public finance and public choice: Analytical perspectives*. New York: Oxford University Press.

Musgrave, R.A., & Musgrave, P.B. (1989). *Public finance in theory and practice*. New York: McGraw Hill Book Company.

Rao, M.G., & Sen, T. K. (1996). *Fiscal federalism in India: Theory and practice*. Macmillan India.

Recommended Readings:

Aronson, J.R. (1985). *Public finance*. New York: McGraw-Hill International.

Basu, K., & Maertens, A. (2012). *The new Oxford companion to economics in India*. New York: Oxford University Press.

Brown, C.V., & Jackson, P. M. (1990). *Public sector economics*. United Kingdom: Wiley-Blackwell.

Buchanan, J. M. (1958). *Public principles of public debt*. Homewood, Illinois: Richard D. Irwin.

Buchanan, J. M., & Musgrave, R. A. (1999). *Public finance and public choice: Two contrasting visions of the state*. Cambridge, Massachusetts: MIT Press.

Chelliah, R.J. (1996). *Towards sustainable growth: Essays in fiscal and financial sector reforms in India*. New York: Oxford University Press.

Datt, R., & Sundaram, K.P.M. (2003). *Indian economy*. New Delhi: S. Chand & Company.

Houghton, R. W. (1970). *Public finance: Selected readings*. Penguin.

Ulbrich, H.H. (2013). *Public finance in theory and practice Second edition*. London: Routledge.

Kurian, N. J. (1999). State government finances: A survey of recent trends. *Economic and Political Weekly*, *34* (19), 1115-1134.

MAECODSE-205 (A)-ADVANCE ISSUES IN POLITICAL ECONOMY

Topology of the code	**Discipline Specific Elective**
Level of the Course	**400**
Course Duration /LTP (Lectures,Tutorials,Practical)	**L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment 20 marks**
Time	**3hours**

Course Objective

The course aims to provide students with a rigorous and systematic understanding of Marxian political economy. It seeks to familiarize them with Marx's methodological approach, including abstraction, dialectical materialism, and historical analysis, and to develop a conceptual grasp of core categories such as commodity, value, money, capital, surplus value, and accumulation. The course emphasizes the theoretical foundations of Marx's critique of capitalism and their relevance for understanding the dynamics and contradictions of capitalist development.

Pedagogy

The course will be conducted primarily through classroom-based teaching, comprising lectures and guided discussions centered on close reading and interpretation of classical Marxian texts. Emphasis will be placed on conceptual clarification, logical reconstruction of arguments, and critical engagement with both primary and secondary literature. Students will be encouraged to relate theoretical constructs to historical processes and contemporary capitalist dynamics through discussions and written assignments.

Course Outcomes

Upon completion of the course, students will be able to critically explain Marx's method of political economy and its philosophical foundations in dialectical materialism. They will develop the capacity to analyze the commodity form, value relations, and the role of money within capitalist exchange, and to interpret the transformation of money into capital and the production of surplus value. Students will also be able to assess the process of capital accumulation, including primitive accumulation, and critically evaluate the historical tendencies and contradictions of capitalist development using Marxian theoretical frameworks.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:

1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question would be compulsory, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to

attempt any 10 short answer type questions. Each short answer type question would carry 2 marks (10x2=20).
4. Rest of the paper shall contain 4 units. Each Unit shall have two questions and the candidates shall be required to attempt on question from each Unit – 4 in all. Each question shall carry 15 marks (15 x 4 = 60).

Unit I
Method of Marxian Political Economy

Marx's Method of Political Economy: the role of abstraction and the movement from the abstract to the concrete.

The historical character of Marx's thought; The three sources and three component parts of Marxism; Dialectical Materialism: from Hegel to Marx

Unit II
Commodity and Money

The commodity form and commodity fetishism; Exchange relations: use value, exchange value, and value

Money and the circulation of commodities

Unit III
The Theory of Capital and Surplus Value

The transformation of money into capital; The general formula for capital; Contradictions in the general formula of capital; Buying and selling of labour-power

The labour process and the process of producing surplus value under capitalism

Unit IV
The Accumulation of Capital

Simple reproduction; Conversion of surplus value into capital; The general law of capitalist accumulation

Primitive accumulation: The secret of primitive accumulation; Expropriation of the agricultural population from the land; Genesis of the industrial capitalist; The historical tendency of capitalist accumulation

Recommended Readings

Unit I

Sweezy, P. M. (1942). *The Theory of Capitalist Development*. Chapters 1 and 2.
Lenin, V. I. (1913). *The Three Sources and Three Component Parts of Marxism*.
Stalin, J. (1938). *Dialectical and Historical Materialism*.
Harvey, D. (2011) *A Companion to Marx's Capital*. Introduction.

Unit II

Marx, K. (1992). *Capital: A Critique of Political Economy*, Vol. I. Chapters 1, 2, and 3.
Harvey, D. (2011). *A Companion to Marx's Capital*. Chapters 1 and 2.

Unit III

Marx, K. (1992). *Capital: A Critique of Political Economy*, Vol. I. Chapters 4, 5, 6, and 7.
Harvey, D. (2011) *A Companion to Marx's Capital*. Chapters 3 and 4.
Sweezy, P. M. (1942). *The Theory of Capitalist Development*. Chapters 1 and 2.

Unit IV

Marx, K. (1992). *Capital: A Critique of Political Economy*, Vol. I. Chapters 23, 24, 25, 26, 27, 31, and 32.

Additional Readings

Shaikh, A. (2016). *Capitalism: Competition, Conflict, Crises*. Oxford University Press.
Lange, O. (1963). *Political Economy*, Vols. I and II.
Nikitin, P. I. (1983). *The Fundamentals of Political Economy*.
Foley, D. K. (1986). *Understanding Capital*. Cambridge, MA: Harvard University Press.
Howard, M. E., Howard, M. C., & King, J. E. (1988). *The Political Economy of Marx*. NYU Press.
Fine, B., & Saad-Filho, A. (2016). *Marx's Capital*. London: Pluto Press.
Robinson, J. (1967). *Essay on Marxian Economics*.
Ollman, B. (2003). *Dance of the Dialectic: Steps in Marx's Method*. University of Illinois Press.
Bottomore, T., Harris, L., Kiernan, V. G., & Miliband, R. (1991). *The Marxist Thought*. Blackwell Publishers Ltd.

MAECODSE-205 (B) - ENVIRONMENTAL ECONOMICS

Topology of the code	**Discipline Specific Core**
Level of the Course	**400**
Course Duration /LTP (Lectures,Tutorials,Practical)	**L=45hours(3 hours aweekfor15weeks=01 semester), T=15hours(1hour a week for 15 weeks=01 semester), P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment 20 marks**
Time	**3hours**

Course Objective: This paper aims to make aware students about the importance of environment in economics and vice-versa. It helps them to know the ways of sustaining our natural resources by optimally allocating them for future use.

Course Outcome: Students will analyze market failures and environmental externalities, determine optimal pollution levels, and evaluate regulatory tools like taxes and permits. They will apply cost-benefit analysis to environmental projects, assess Indian environmental laws and EIA, and discuss global issues and sustainable development strategies

Pedagogy of the Course Work: The course relies on a combination of lectures, solving problems, and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. Special tutorials/contact hour for one-to-one student-teacher interactions.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:

1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question, which would be compulsory, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks ($10 \times 2 = 20$).
4. Rest of the paper shall contain 4 units. Each unit shall have two questions and the candidates shall be required to attempt one question from each Unit— 4 in all. Each question shall carry 15 marks ($15 \times 4 = 60$).

UNIT – I

Introduction: Basic concepts of Environment, Economics and Ecology. Relation betweenEnvironment and Economy (inter linkages), the natural environment and the human economy-theneoclassical economics (introductory) and first two laws of thermodynamic under ecologicalperspective.

Pollution and resource degradation as externalities: the open access and the public good problem; common property resources and economic problem.

UNIT- II

Economics of pollution: Damage and abatement cost functions; Marginal decision; Optimum levelof pollution; Property Rights or Coasian Theorem; role of state in environment preservation-reviewof environment legislation in India.

UNIT- III

Cost-Benefit Analysis and Environment: measuring the costs and benefits of environmentprotection. Environment regulations and legislations- Air, Water, Land, chemicals and pesticides;
Understanding Environment Impact Assessment (EIA), Green National Accounts andEnvironmental Accounting.

UNIT- IV

Global Environmental Pollution: acid rain, ozone depletion and global warming (causes,consequences and solutions or international responses); Earth Summits 1992 and 2002.

Economic growth and development; the environmental Kuznets curve; Economics of sustainable development; Trade, development and environment under WTO regime.

Essential Readings

Eugine, T. (2005). *Environmental Economics*. Delhi: Varinda Publications.

Goodstein, E.S. (2002). *Economics and the Environment* (3rd ed.). John Wiley & Sons. Inc.

Hanley, N., Shorgen, J. F. & Ben, W. (2001). *Introduction to Environmental Economics*. Oxford University Press.

Hussen, A. (2004). *Principles of Environmental Economics* (2nd ed.). London & New York: Routledge, Taylore & Francis Group.

Kolstad. C.D. (2011). *Intermediate Environmental Economics(second edition).*Oxford University Press Inc.

Nandimath, O.V. (2009). *Handbook of Environmental Decision Making in India*. Oxford University Press.

Further Readings

Bhattachrya, R. N. (2001). *Environmental Economics-An Indian Perspective*. Oxford University Press.

Biswas, M. (2007). *Environmental Economics*. New Delhi: Mittal Publications.

Karpagam, M. (1991). *Environmental Economics*. New Delhi: Sterling Publishers.

Sankar, U. (2001). *Environmental Economics*. Oxford University Press.

Singh, K. & Shishodia, A. (2007). *Environmental Economics: Theory and Applications*. Sage Publication.

Tietenberg, T. (2003). *Environmental and Natural Resource Economics,*(6th ed.). Pearson Education.

MAECODSE-205 (C) – ECONOMICS OF MONEY AND BANKING

Topology of the code	**Discipline Specific Core**
Level of the Course	**400**
Course Duration /LTP (Lectures,Tutorials,Practical)	**L=45hours (3 hours aweekfor15weeks=01 semester), T=15hours (1hour a week for 15 weeks=01 semester), P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment 20 marks**
Time	**3hours**

Objectives: Money and banking constitute important components towards understanding of economics. A clear understanding of the operations of money and banking and their interaction with the rest of the economy is essential to realize how monetary forces operate through a multitude of channels – market, non-market, institutions and among others, the state. The operation of financial markets and their regulations are to be studied to appreciate their key-role in an economy, especially after the far reaching banking and financial sector reforms in India and elsewhere. The present course is designed to acquaint the students fully with the changing role of financial institutions in the process of growth and development. Accordingly, the paper on 'Economics of Money and Banking' is an optimal integration of monetary theory, banking and non-banking financial institutions, which combine with itself a systematic discussion of the theory, institutions and policy with special reference to India.

The students would be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Pedagogy of the Course Work:

The course relies on a combination of lectures, solving problems, and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. The students would be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The script would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Course outcomes:

Upon completion of the course, the student will be able to analyze the creation, components, and measurement of money, apply theories of money demand and liquidity, and evaluate the term structure of interest rates in the Indian context; examine the objectives, tools, and transmission mechanisms of monetary policy and critically assess the functioning of the RBI, commercial banks, and banking reforms; evaluate the role, growth, and regulation of non-banking financial institutions; and analyze the structure and functioning of the international monetary system, including the IMF, the European Monetary System, and global money and capital markets.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:

1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question would be compulsory, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks (10x2=20).
4. Rest of the paper shall contain 4 units. Each Unit shall have two questions and the candidates shall be required to attempt on question from each Unit – 4 in all. Each question shall carry 15 marks (15 x 4 = 60).
5. Candidates are allowed to use Simple, Non-Scientific and Non-Programmable Calculator in the Examination Hall.
6. It is compulsory for the paper setter to mention the requirement of log, antilog and statistical tables.

UNIT-I

Money Supply: Theoretical and Empirical Attempts to Define Money. Balance sheet of a Central Bank and Components of Money Supply. Money Creation by the Banking System. High Powered Money and Money Multiplier. Measures of Money Supply and liquidity in India.

Demand for Money: Interest sensitivity of demand for money- A review of Classical, Keynesian and Monetarist theories of demand for Money. Empirical attempts to estimate demand for money function.

UNIT-II

Term Structure of Interest Rates: The Expectations Theory, Liquidity Premium Theory, Market Segmented Theory, Preferred Habitat Theory, Structure of Interest Rates in India.

Monetary Policy: Targets, Goals and the Trade Offs among Alternate Goals, Lags in Operation, Keynesian and Monetarist Views on Transmission Mechanism, Rules vs. Discretion.

UNIT-III

Central Banking: Main Functions, Policy Tools, Recent Monetary Policy of RBI, money aggregates targeting, interest targeting and inflation targeting approaches of RBI, Autonomy of RBI.

Commercial Banking: Types of commercial banks in India. Banking Sector Reforms, Major Developments in Commercial Banking in India since reforms (including mergers).

UNIT-IV

Non-Banking Financial Institutions: Role, Growth and Structure of Non-Banking Financial Institutions (NBFIs) in India. Types and Control of Non-Banking Financial Companies (NBFCs).
International Monetary System: IMF as provider of International Liquidity. European Monetary System–Features of the Maastricht Treaty and Post Maastricht Treaty Developments. Constituents of international money and capital markets.

Recommended Readings:

Bhole, L.M. (2016). Financial Institutions and Markets. Tata McGraw Hill.

Bofinger, P. (2001). Monetary Policy: Goals, Institutions, Strategy and Instruments. Oxford University Press.

Dennis Geoggrey, E.J. (1981). Monetary Economics. Longman.

Desai, V. (2019). The Indian Financial System and development. Himalaya Publishing House.

Diulio, A. E. (1987). Theory and Problems of Money and Banking. International Edition (Schaum's Outline Series).

Fabozzi, F. J. & Franco (2009). Capital Markets Institutions and Instruments, (4th ed.). New Jersey: Prentice Hall, Modigliani.

Gordon, E. & Natarajan, K. (2019), Financial Markets & Services. Himalaya Publishing House.

Howells, P. & Bain, K. (2002). The Economics of Money, Banking and Finance. Prentice Hall, Pearson Education Ltd.

Khan, M.Y. (2017). Indian Financial System. Tata Mcgraw Hill.

Kulkarni, G. (1999). Modern Monetary Theory. New Delhi: Macmillan.

Mankiw, N.G. (2008). Macro-economics. New York: Macmillan Worth Publishers.

Mohan, R. (2009). Monetary Policy in a Globalized Economy – A Practitioner's View. Oxford University Press. Government of India Economic Survey (Recent Issues).

Reserve Bank of India Various Reports.

SEMESTER – III

Level of Course	M.A. Economics with Research		
	Course Code	**Course Name**	**Credits**
L-500	MAECORBC-RM-301 (A)	Research Methodology	4
	MAECODSC-302	Economics of Growth, Development and Planning-I	4
	MAECODSC-303	Indian Economy-I	4
	Any One Paper from: MAECODSE-304 (A) or MAECODSE-304 (B) or MAECODSE-304 (C) or MAECODSE-304 (D)	**Any one Papers from:** Econometrics-I or Economics of Agriculture-I or Economics of Industry-I or Mathematical Economics-I	4
	MAECORBC-DST-305 (P1)	Dissertation	4

MAECORBC-RM-301(A): RESEARCH METHODOLOGY

Topologyofthecode	**Research Based Course**
LeveloftheCourse	**500**
CourseDuration/LTP(Lectures, Tutorials, Practical)	**L=60 hours(4 hours aweekfor15weeks=01 semester), T=00 hours, P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment20 marks**
Time	**3 hours**

Objectives:

To rigorously train the students with the concepts, methodology and reasoning involved in analyzing the economic behavior of firms and markets, in general, a static and partial equilibrium frameworks.

Pedagogy of the Course Work:

The time–tested method of imparting verbal instructions through lectures would be used. Examples, in so far as possible, would be selected from everyday life/experience. The feedback would be sought through the students' written assignments, tutorials and seminars. The students

would be evaluated at the end of the academic year through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Course outcomes:

After completing this course, students will acquire a comprehensive understanding of research methodology and the scientific research process in social sciences. They will be able to formulate research problems, review and cite literature using standard tools, and prepare research synopses, reports, and theses. The course will develop students' competence in selecting appropriate data collection, measurement, and scaling techniques under quantitative and qualitative approaches. Students will also gain the ability to apply parametric and non-parametric statistical tests for hypothesis testing and critically interpret research findings.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:

1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question would be **compulsory** and shall be a short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks (10×2 = 20).
4. The rest of the paper shall contain four units. Each unit shall have two questions and the candidates shall be required to attempt one question from each unit – 4 in all. Each question shall carry 15 marks (15×4 = 60).
5. Candidates are allowed to use Simple, Non-Scientific and Non-Programmable Calculator in the Examination Hall.
6. It is compulsory for the paper setter to mention the requirement of log, antilog and statistical tables.

UNIT – I

Introduction to Research Methodology: Meaning and Objectives of Research, Types of Research, Defining the Research Problem. Reviewing the Literature, Using Mendeley in Citing the Literature.

Triangulation: Concept, Rationale and Process.

Meta Analysis: Concept, Rationale and Process.

UNIT – II

Inter-disciplinary Research in Social Sciences: Rationale and Approaches.

Social Science Research Process. Selection and Formulation of Research Problem under Quantitative and Qualitative research approach.

Research Synopsis: Concept, Format and Writing. Report and Thesis Writing.

UNIT – III

Data Collection Techniques: Questionnaire Method: Meaning, Types, Format and Pre-Testing of Questionnaires. Interview Method: Meaning, Types, Process and Rules of Interviews. Observation Method: Meaning and Types.

Measurement and Scaling Techniques: Measurement Scales: Types of Measurement Scales, Types and Components of Measurement. Accuracy of Measurement and Reliability Testing.

UNIT – IV

Testing of Hypotheses: Parametric Tests: Applications of t, z, and F-test Statistics. Non-Parametric Tests: Application of Chi-square, Mann-Whitney and Kruskal-Wallis Tests.
Data Sources of Indian Economy: Nature and Sources of Micro and Macro statistical data.

Recommended Readings:

Goon, P : An Outline of Statistical Theory, World Press Pvt. Ltd., Calcutta.
Kumar Pranesh, Singh Daroga, Singh Padam: Handbook of Sampling Methods, Indian and Agricultural Statistics Research Institute, ICAR.
Anderson, T. W.(1952): An Introduction to Multivariate Statistical.
Goode, W J and Hatt, Paul K (1952): Methods in Social Research. McGraw Hill Book Company, New York.
Sukhatme and Sukwatme (1970): Sampling Theory of Surveys with Applications. John Wiley and Sons.
Tripathi, P C. (2008) : A Textbook of Research Methodology, Sultan Chand.
Kumar, Ranjit (2009) : Research Methodology: Methods and Techniques, New Age International, India.
Creswell, J.W. (1994). Research Design: Qualitative and Quantitative Approaches, London, Sage Publications.
Miah, Abdul Quader. (2016). Applied Statistics for Social and Management Sciences, (1st ed.) Springer, Singapore.
Moser. C.A. and Kalton G. (1979). Survey Methods in Social Investigation, (2nd ed.) The English Language Book Society and Heinemann Educational Books. London.
Delbert C. Miller. Handbook of Research Design and Social Measurement, (5th ed.) Sage Publications, London.
David Nachmias and Chara Nachmias. Research Methods in the Social Sciences. (2nd ed.) St. Martin's Press Inc, New York.
Clarie S.L. Research Methods in Social Relations, Holt Rinehart and Winston. New York.
Pauline V. Young. (2007) Scientific Social Research and Surveys, India. (4th ed.) Asia Publishing House.
William J. Goode and Paul Hatt. (2006) Methods in Social Research, (Ist Indian Reprint). USA, Mc Graw K. Hill Book Company.
Norman K. Denzin, Yvonna Lincoln. (2003). The Landscape of Qualitative Research: Theories and Issues. Sage Publications, New Delhi.
Clive Seale, Giampietro Gobo, Jaber I. Gubrium & David Silverman.(2004) Qualitative Research Practice. Sage Publications, New Delhi.
Dawn Durton. (2000). Research Training for Social Scientists, Sage Publications, New Delhi.
Royle A. Singleton, Jr. and Bruce C. Straits. (1999). Approaches to Social Science Research, Oxford University Press.
Mason, Jennifer. (2002). Qualitative Researching. Sage.
Chandrasekhar C.P.& Jandhyala B.G. Tilak (2001) India's Social-Economic Database – surveys of selected areas, Tulika, New Delhi.

Alan Bryman (2015). Social Research Methods. (5thed.) Oxford University Press, United Kingdom.

MAECODSC-302: ECONOMICS OF GROWTH, DEVELOPMENT AND PLANNING – I

Topology of the code	**Discipline Specific Core**
Level of the Course	**500**
CourseDuration/LTP(Lectures, Tutorials, Practical)	**L=60 hours(4 hours aweekfor15weeks=01 semester), T=00 hours, P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment20 marks**
Time	**3 hours**

Objectives:

As a sequel to the post-second war developments, the study of Economic Development gained impetus because three-fourths of humanity was experiencing wretched conditions of existence. There was a pressing need in those countries for uplifting their economic conditions by restructuring their economies to acquire greater diversity, efficiency and equity, in consonance with their priorities. Since a variety of perspectives were available, the policy makers were eager to acquaint themselves with various policy options in their bid to re-construct their dilapidated economies. In addition, various international bodies were also keen to help and guide the laggards. Consequently, the study of development economics assumed greater significance. In recent times, the resurgence of marketism and greater focus on areas like education, health, sanitation, energy and environment, and infrastructure development, hitherto relegated to the background, have reopened some of the old debates besides opening up new areas of investigation.

Growth and Development economists are making earnest efforts at theorizing to break fresh grounds. Consequently, study of this discipline continues to be of prime importance. Modules incorporated in this paper are devoted to the theories of growth and development, importance of agriculture, and the rational and pattern of industrialization in developing countries. The other important issues in the context of development such as infrastructure – linkages, role of international trade, importance of economic policies and relevance of planning have been included in the modules of this paper.

Pedagogy of the Course Work:

The time–tested method of imparting verbal instructions through lectures would be used. Examples, in so far as possible, would be selected from everyday life/experience. The feedback would be sought through the students' written assignments, tutorials and seminars. The students would be evaluated at the end of the academic year through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Course outcomes:

Upon successful completion of this course, students will be able to critically understand the concept of economic development and its evolving dimensions, including inequality, human development, and sustainability. They will gain analytical knowledge of the structural

characteristics of developing economies and major development theories and models. The course will enable students to evaluate population dynamics, institutional roles, environmental concerns, and global dependency issues in the development process. Students will also develop the ability to apply theoretical frameworks to real-world development problems and formulate informed policy perspectives relevant to developing nations.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:

1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question would be compulsory, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks (10x2=20).
4. Rest of the paper shall contain 4 units. Each Unit shall have two questions and the candidates shall be required to attempt on question from each Unit – 4 in all. Each question shall carry 15 marks (15 x 4 = 60).
5. Candidates are allowed to use Simple, Non-Scientific and Non-Programmable Calculator in the Examination Hall.
6. It is compulsory for the paper setter to mention the requirement of log, antilog and statistical tables.

UNIT – I

Understanding Development: Measuring Inequalities in a heterogeneous World – Islands of Prosperity and how the other half Lives. Dividing the World and levels of development. Development as an evolving concept.

Goulet's Three Core Values of Development. Sen's Conception of Development.

Income based Measures and their Inadequacies. PQLI and HDI as indicators of development.

UNIT-II

Common Characteristics of Developing Nations: The Vicious Circle of Poverty (Nurkse), Low Level Equilibrium Trap (Nelson), Critical Minimum Effort Theory (Leibenstein). Dualism (Social and Technological). Institutions and Economic Development (D.C. North). Meaning of Dualism, its Characteristics and forms.

Development in Historical Perspective: Dependency theory: its forms and effects. Neo Imperialism and Neo colonialism.

UNIT – III

Theories and Models of Development: Classical, Karl Marx, Schumpeter, Rostow's Theory of stages of Economic growth. Harrod-Domar Model, Solow Model, Endogenous growth models.

UNIT – IV

Population Growth, Economic Development and Environment: Theory of Demographic Transition, Interrelation between Population Growth and Economic Development, Urban Development and Environment. Natural resources, Environmental degradation and Sustainable Economic Development.

Recommended Readings:

Agarwala, A. N., & Singh, S. P. (Eds.). (1963). *Economics of underdevelopment: A series of articles and papers*. London: Oxford University Press.

Cypher, J. (2014). *The process of economic development*. London: Routledge.

Lucas Jr, R. E. (1988). On the mechanics of economic development. *Journal of Monetary Economics*, 22(1), 3-42.

Meier, G. M., & Rauch, J. E. (Eds.). (1995). *Leading issues in economic development (Vol. 6)*. New York: Oxford University Press.

Ray, D. (1998). *Development economics*. New Jersey: Princeton University Press.

Romer, P. M. (1986). Increasing returns and long-run growth. *Journal of Political Economy*, 94(5), 1002-1037.

Romer, P. M. (1990). Endogenous technological change. *Journal of Political Economy*, 98(5, Part 2), S71-S102.

Schultz, T. W. (1982). *Investing in people: The economics of population quality*. USA: University of California Press.

Sen, A. (1999). *Development as freedom*. New Delhi : Oxford University Press.

Thirlwall, A.P. (Latest edition). *Economics of development: Theory and evidence*. New York: Palgrave Macmillan.

Todaro, M. P., & Stephen, C. S. (Latest edition). *Economic Development*. New York: Addison Wasley Logman.

MAECODSC - 303: INDIAN ECONOMY–I

Topology of the code	**Discipline Specific Core**
Level of the Course	**500**
CourseDuration/LTP(Lectures, Tutorials, Practical)	**L=60 hours(4 hours aweekfor15weeks=01 semester), T=00 hours, P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment20 marks**
Time	**3 hours**

Objectives:

The objective of this paper is to acquaint the students with the performance of different sectors of the Indian economy and the policy framework governing them. This will provide them an insight into the past, present and future functioning of the Indian economy and strengthen their analytical capability. The students would be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The script would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Pedagogy of the Course Work:
The time–tested method of imparting verbal instructions through lectures would be used. Examples, in so far as possible, would be selected from everyday life/experience. The feedback would be sought through the students' written assignments, tutorials and seminars. The students would be evaluated at the end of the academic year through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Course outcomes:
Upon completion of this course, students will be able to analyze the historical evolution of the Indian economy from the colonial period to the post-reform era. They will understand the structure and dynamics of labour force growth, poverty, unemployment, and regional disparities in India. Students will be able to assess the performance of Indian agriculture and industry in the context of economic reforms. The course will also enable students to critically evaluate policy initiatives and sectoral growth patterns shaping India's development trajectory.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:
1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question would be compulsory, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks (10x2=20).
4. Rest of the paper shall contain 4 units. Each Unit shall have two questions and the candidates shall be required to attempt on question from each Unit – 4 in all. Each question shall carry 15 marks (15 x 4 = 60).
5. Candidates are allowed to use Simple, Non-Scientific and Non-Programmable Calculator in the Examination Hall.
6. It is compulsory for the paper setter to mention the requirement of log, antilog and statistical tables.

UNIT- I

Indian Economy in Pre-Reforms Era:
Economic Consequences of British Rule in India (1850-1947).
Features of Indian Economy during Pre-Reforms Era (1947-1991).
State of Indian Economy in 1990-91: Features and Appraisal of Economic Reforms Programme.

UNIT- II

Labour Force, Poverty and Unemployment in India:
Labour Force Growth and Occupational Structure since Economic Reforms.
Poverty and Unemployment: Nature, Extent and Estimates and Policy Initiatives since Economic Reforms.
Inter-state Disparities in the Pattern of Development in the Post-Reform Period.

UNIT- III

Performance of Indian Agriculture:

Agriculture: Pattern of Growth of Indian Agriculture since 1950. Economic Reforms and Agricultural Growth in India. Regional Imbalances in Indian Agriculture.

Secondary Agriculture: Need, Potential and Role in India. Food Security – Problems and Policy Options.

UNIT- IV

Performance of Indian Industry:

Industry: Trends in Growth and Structure of Indian Industry since 1950. Industrial Policy Initiatives during Post-Reform Period. Economic Reforms and Manufacturing Sector Growth in India.

Recommended Readings:

Books:

Acharya, S. & Rakesh, M. (2011). India's Economic Performance and Challenges: Essay's in Honour of Montek S Ahluwalia. New Delhi: Oxford University Press.

Ahluwalia, I. J. & Little, IMD. (2008). India's Economic Reforms & Development (Essays in Honour of Manmohan Singh). New Delhi: Oxford University Press.

Datt, G & Ashwani, M. (Latest Edition). Indian Economy. New Delhi: S.Chand.

Dhar, P.K. (Latest Edition) Indian Economy-Its Growing Dimensions. Kalyani Publisher.

Fernando, A.C. (2016). Indian Economy. India: Pearson.

Kapila, U. (Latest Edition) Indian Economy since Independence. New Delhi: Academic Foundation.

Kapila, U. (Latest Edition) Indian Economy: Performance and Policies. New Delhi: Academic Foundation.

Krueger, A.O. (2002). Economic Policy Reforms and the Indian Economy. New Delhi: Oxford University Press.

Kumar, D & Raychaudhary, T. (Eds) (1983) The Cambridge Economic History of India (1751-1970), Vol. 2, Cambridge University Press.

Mishra, S.K. & Puri, V.K. (Latest Edition). Indian Economy. Mumbai: Himalya Publisher House.

Roy, T. Economic History of India (1857-1947), (3rd ed.). India: Oxford University Press.

Economic and Political Weekly, Various Issues.

Government of India, Economic Survey (Various Issues). New Delhi: Oxford University Press.

Ahluwalia, M.S. (2002). Economic Reforms in India since 1991: Has Gradualism worked? Journal ofEconomic Perspectives, summer, 16(3), 67–88.

Bhagwati, J & Arvind P. (2012). India's Tryst with Destiny. India Harper Collins.

Bhalla, G.S. & Gurmail Singh. (2009). Economic Liberalisation and Indian Agriculture: A State wise
Analysis. Economic & Political Weekly, 44(52), 34–44.
Dreze J. & Sen, A. (2013). An uncertain Glory-India & its Contradictions. New Delhi: Penguin Books.
Essays from EPW-Quarter Century of Liberalisation in India (2018). Oxford University Press.

Research Papers:

Himanshu. (2011). Employment Trend in India: A Re-Examination. Economic & Political Weekly, XLVI (37), 43-59.
Mani, S. (2011). National Manufacturing Policy: Making India a Power House. Economic & Political Weekly, XLVI (53), 16-19.
Mehrotra,S., Gandhi, A., Saha, P & B. K. S. (2013). Turnaround in India's Employment Story Silver
Lining amidst Joblessness and Informalisation? Economic and Political Weekly, XLVIII (35), 87–96.
Nayyar, G. (2008). Economic Growth and Regional Inequality in India. Economic & Political Weekly, 43(06).
Panagariya, A. (2004). Growth and Reforms during 1980's and 1990's. Economic & Political Weekly,29 (25), 2581-2594.
Paranjape, H.K. (1991). New Industrial Policy: A Capitalist Manifesto. Economic & Political Weekly, 26 (43).
Patnaik, U. (2013). Poverty Trends in India 2004-05 to 2009-10. Economic and Political Weekly, XLVIII (40), 43-58.
Singh, M. (2018). Changing India, Vol. 1-5. Oxford University Press.

MAECODSE-304 (A): ECONOMETRICS-I

Topology of the code	**Discipline Specific Elective**
Level	**500**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L=60 hours(4 hours aweekfor15weeks=01 semester), T=00 hours, P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment20 marks**
Time	**3hours**

Objectives:

Applications of economic theory need a reasonable understanding of economic relationships and relevant statistical methods. The econometric theory thus becomes a very powerful tool for understanding of applied economic relationships and for meaningful research in economics. This paper accordingly is devoted to equip the students with basic theory of

econometrics and relevant applications of the methods. The topics covered in the course include various problems faced in estimation of both single equations and simultaneous equation models.

Besides the time-tested method of imparting verbal instructions through lectures, description and derivation of econometric models would be demonstrated by extensive use of blackboard. Examples, in so far as possible, would be selected from everyday life/experience. The feedback would be sought through the students writing assignments, tutorials and seminars.

The students would be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Pedagogy of the Course Work:

The course relies on a combination of lectures, solving problems, and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. The students would be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The script would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Course outcomes:

Upon completion of this course, students will be able to understand the nature, scope, and methodology of econometrics and apply econometric techniques to analyse economic relationships. They will gain proficiency in estimating simple and multiple regression models using OLS and MLE, and in testing hypotheses while understanding the properties of estimators. Students will be able to detect, interpret, and correct econometric problems such as multicollinearity, heteroscedasticity, and autocorrelation, and address issues of model specification and measurement errors. The course will also equip students with practical skills in using statistical software (SPSS, E-Views, STATA) for estimation, diagnostic testing, and policy-oriented interpretation of econometric models.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:

1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question would be compulsory, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks (10x2=20).
4. Rest of the paper shall contain 4 units. Each Unit shall have two questions and the candidates shall be required to attempt on question from each Unit – 4 in all. Each question shall carry 15 marks (15 x 4 = 60).
5. Candidates are allowed to use Simple, Non-Scientific and Non-Programmable Calculator in the Examination Hall.
6. It is compulsory for the paper setter to mention the requirement of log, antilog and statistical tables.

UNIT–I

Nature& Types of Econometrics, Methodology Followed& Type of Data Used in Econometric Models. Estimation of the Two Variable and Multiple Regression Models with the Method of Ordinary Least Squares (OLS); BLUE Properties and Testing of Hypothesis, Maximum Likelihood Estimation (MLE).

UNIT-II

Nature, Consequences, Detection & Remedial Measures for the problems of: Multicollinearity, Hetroscedasticity and Autocorrelation.

UNIT-III

Model Specification and Diagnostic Testing: Types of Specification Errors and consequences, Tests of Specification and Misspecification Errors of Measurement in dependent and independent variables Encompassing and Non-Encompassing Models: Introduction Criteria for Model Selection.

UNIT-IV

Estimation of Regression Models using Standard Statistical/Econometric Packages [SPSS/E-View/STATA]. Interpretation of Estimates and Hypothesis Testing. Testing for and redressal of Multicollinearity, Hetroscedasticity, Autocorrelation using Computer Techniques.

Recommended Readings:

Gujarati, D. (2002). Basic Econometrics (4th ed.). McGraw Hill.

Stock, J.H. & Watson, M. W. (2011). Introduction to Econometrics (3rd ed.). Delhi: Pearson

Prentice Pvt. Ltd.

Johnston, J. & Nardo, D. (1997). Econometric Methods (4th ed.). McGraw Hill.

Kemanta, J. (2008). Elements of Econometrics (2nd ed.). New York: Macmillan.

Koutsoyiannis, A. (1973). A Theory of Econometrics. New York: Harper & Row.

Maddala, G. S. (2007). Introduction to Econometrics (3rd ed.). India: Wiley.

Wooldridge, J. M. (2000). Introductory Econometrics. South Western College Publishing.

MAECODSE-304 (B): ECONOMICS OF AGRICULTURE-I

Topologyofthecode	**Discipline Specific Elective**
Level	**500**
CourseDuration/LTP(Lectures, Tutorials, Practical)	**L=60 hours(4 hours aweekfor15weeks=01 semester), T=00 hours, P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment20 marks**
Time	**3hours**

Objective:
The focus of this course is to introduce students to the importance of agricultural sector and to the issues associated with agricultural transformation, agricultural production, and demand of and supply for agricultural commodities. The objective is to equip students to analyze and critically assess issues, policies and programmers in these areas with particular emphasis on Indian agriculture.
The students would be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/options concerned.

Pedagogy of the Course Work:
The course relies on a combination of lectures, solving problems, and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. The students would be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The script would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Course outcomes:
This course provides students with a comprehensive understanding of agricultural economics, covering the role of agriculture in economic development, farm production, and inter-sector linkages. Students will analyze agricultural finance, credit markets, risk management, and policies, as well as the structure, efficiency, and stabilization of agricultural markets. The course examines theories and models of agricultural transformation, including Indian and international experiences, and equips students to evaluate demand-supply dynamics, trade, and food security issues. Overall, it enables students to assess agricultural development strategies, emerging challenges, and policy interventions for sustainable growth.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:
1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question would be compulsory, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks (10x2=20).
4. Rest of the paper shall contain 4 units. Each Unit shall have two questions and the candidates shall be required to attempt on question from each Unit – 4 in all. Each question shall carry 15 marks (15 x 4 = 60).
5. Candidates are allowed to use Simple, Non-Scientific and Non-Programmable Calculator in the Examination Hall.
6. It is compulsory for the paper setter to mention the requirement of log, antilog and statistical tables.

UNIT-I

Introduction: Nature and Scope of Economics of Agriculture; Specificities of Farm Organization and Agricultural Production and Markets; Inter-sector Linkages of Agriculture [Backward and Forward Linkages].

Role of Agriculture in Economic Development: Contribution of Agriculture to Economic Development; Historical Perspective [Experience of England, Japan, and India]; Declining Role of Agriculture and Dilemmas of Development.

UNIT-II

Agricultural Production: Uniqueness of Agricultural Production; Various Types of Factor-Product, Factor-Factor and Product-Product Relations; Cobb-Douglas and CES Production Functions.

Role of Farm Size and Structure in Equilibrium, Productivity, and Return to Scale. Sources and Barriers to Agricultural Development.

UNIT-III

Agricultural Transformation: Schultz Theory of Transformation of Traditional Agriculture Mellor's Model of Agricultural Development; Hayami- Ruttan Induced Innovation Model of Agricultural Development.

Agricultural Transformation: Indian Experience.

UNIT-IV

Risk and Uncertainty in Agriculture: Nature and Types of Risks and Uncertainties in Agriculture, Public Policies and Farm Level Measures to Reduce Risk and Uncertainties.

Demand for Farm Products: Characteristics of Demand for Farm Products; Quantity and Quality Components of Demand for Food; Forecasting of Demand for Food Products.

Supply of Farm Products: Characteristics of Supply of Farm Products, Supply Response of Farm Products; Role of Price and Non-Price Factors in Inducing Aggregate Supply, Empirical Supply Response Functions and Evidence.

Recommended Readings:

Bruce L. G. & Gordon C. R. (2002). Handbook of Agricultural Economics Vol.2A, Agricultural and Its External Linkages. Amersterdam, Elsevier Science B.V.
Bruce L. G. & Gordon C. R. (2002). Handbook of Agricultural Economics Vol. 2B, Agricultural and Policy. Amersterdam, Elsevier Science B.V.
Bruce L. G. & Gordon C. R. (2002). Handbook of Agricultural Economics Vol.1A, Agricultural Production. Amersterdam, Elsevier Science B.V.
Bruce L. G. & Gordon C. R. (2002). Handbook of Agricultural Economics Vol.1B, Marketing Distribution and Consumption. Amersterdam, Elsevier Science B.V.
Karl, E. & Witt, L. (1964). Agriculture and Economic Development. New York: McGraw Hill Company.

Southworth, H. & Johnston, B. (2011). Agricultural Development and Economic Growth. New York : Cornell University Press.
Subrata, G. & Ingersent, K. (1984). Agriculture and Economic Development. London : Harvestor Press Ltd.
Tsakok, I. (2011). Success in Agricultural Transformation: What It Means and What Makes It Happen. New York: Cambridge University Press.

MAECODSE-304 (C): ECONOMICS OF INDUSTRY-I

Topology of the code	**Discipline Specific Elective**
Level	**500**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L=60 hours(4 hours aweekfor15weeks=01 semester), T=00 hours, P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment20 marks**
Time	**3hours**

Objectives:

In the contemporary world with globalization and liberalization, more and more attention is being given to industry. Since industry performance critically depends on firms' behaviour allowing equilibrium outcome, the course intends to provide a rigorous knowledge of different long-run equilibrium out comeof firms under different conditions from the point of view of public policy. The students are also equipped to deal with debates involved in the industrial development in a cogent and analytical manner, particularly in the Indian context. However, it should be noted that Game Theoretic approach to any topic/problem is outside the scope of the present syllabus.

The students would be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Pedagogy of the Course Work:

The course relies on a combination of lectures, solving problems, and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. The students would be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The script would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Course outcomes:

This course provides students with a comprehensive understanding of industrial organization, market structures, and firm behaviour. Students will analyse profit and non-profit maximization hypotheses, monopoly and oligopoly power, strategic conduct under interdependence, and the role of innovation in competitive outcomes. The course covers mergers, takeovers, pricing strategies, price discrimination, and efficiency implications, along with the evaluation of market performance and public policy interventions. By integrating theoretical models with empirical and policy perspectives, students will be equipped to critically assess firm strategy, market efficiency, and the evolution of regulatory frameworks, including the Indian context.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:

1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question would be compulsory, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks (10x2=20).
4. Rest of the paper shall contain 4 units. Each Unit shall have two questions and the candidates shall be required to attempt on question from each Unit – 4 in all. Each question shall carry 15 marks (15 x 4 = 60).
5. Candidates are allowed to use Simple, Non-Scientific and Non-Programmable Calculator in the Examination Hall.
6. It is compulsory for the paper setter to mention the requirement of log, antilog and statistical tables.

UNIT – I

Constraints of Proprietorship. Advantages of Modern Corporation. Critiques of Profit Maximization
Hypothesis.
Non-profit Maximization models: Boumol, Williamson, Marris and Cyert and March. Critical overview of Non-Profit Maximization Hypothesis

UNIT – II

Monopoly Power and Oligopolistic Market Structure. Measures of Sellers' Concentration and advantagesof the HHI index.
Deterministic Explanation of Sellers' Concentration: Economies of Scale, Barriers to Entry, Mergers, Size and Growth of Markets; Stochastic Explanation

UNIT – III

Market Conduct under Oligopoly: Concepts and Overview of Outcomes under Interdependence: Concepts of Cournot and Bertrand Rivalry, Collusive Conduct and Dominant Firm Behaviour and Potential Competition. Limit Price and Contestable Markets.

Non-Price Competition with Reference toAdvertising: Dorfman-Steiner Condition and its Critique. Evolution of Structure-Conduct-PerformanceHypothesis.

UNIT- IV

Market Performance: Market Structure and Profitability; Collusion versus Efficiency. Issue of AllocativeEfficiency.
Issues of Productive Efficiency and Sub-Optimal Capacity; factors explaining sub-optimalcapacity.

Recommended Readings:

Bain, J. S. (1968). Industrial Organizations (2nd ed.). University of California.
Cohen, K. L. & Cyert R. M. (1976). Theory of the Firm. Prentice Hall.
Devine, P. J. & Others (1976). Introduction to Industrial Economics. George Allen & Unwin.
Hawkin, C. J. (1973). Theory of the Firm. Macmillan.
Hay, D. A. & Morris, D. J. (1991). Industrial Economics: Theory and Evidence. Oxford University Press.
Kansra, S. L. (1990). Monopoly Power: Some Aspects of its Impact on Indian Industry (Unpublished Ph.D. Thesis, P. U.).
Martin, S. (1989). Industrial Economics: Economic Analysis and Public Policy.
Scherer, F. M. & Ross, D. (1990). Industrial Market Structure and Economic Performance, Boston Houghton Miffilin Company.
Shepherd, W. G. (1990). The Economics of Industrial Organisation. New Jersey: Prentice Hall Englewood Cliffs.
Wildsmith, S. (1975). Managerial Theories of the Firm.

MAECODSE-304 (D): MATHEMATICAL ECONOMICS-I

Topologyofthecode	**Discipline Specific Elective**
Level	**500**
CourseDuration/LTP(Lectures, Tutorials, Practical)	**L=60 hours(4 hours aweekfor15weeks=01 semester), T=00 hours, P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment20 marks**
Time	**3hours**

Objectives:

This course is designed to equip students to understand the economic concepts and theories by using mathematical tools and techniques to refine the verbal logic. The use of calculus has permitted formulation of economic problems in multivariable mode and yield valuable insight about optimizing human behaviour. Modern algebraic tools allow convenient handling of simultaneous equations in the context of linear programming, game theory and input-output analysis. Mathematical economics deals with various applications of mathematical tools and techniques in defining and developing economic relationships. This course has been accordingly designed to include various mathematical techniques/methods/models related to the different parts of economic theory like consumer theory, theory of production, pricing, trade cycles, growth models etc. The paper covers important aspects of microeconomics; macroeconomics and development theory and the various modules have been accordingly designed. The students would be evaluated at the end of each semester through subjective type

questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Pedagogy of the Course Work:
The course relies on a combination of lectures, solving problems, and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. The students would be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The script would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Course outcomes:
Upon completion of this course, students will be able to apply optimization techniques, including linear and non-linear programming, to measure efficiency, solve economic allocation problems, and address issues like peak load pricing. They will understand dynamic optimization methods, including calculus of variations, control problems, and Pontryagin's maximum principle, and construct and interpret phase diagrams. Students will be able to model and analyse economic systems using static and dynamic input–output frameworks, including activity analysis, cost and price determination, and social accounting matrices. The course will also equip students with game theory tools to analyse strategic interactions, compute Nash equilibria in static and dynamic games, and interpret real-world scenarios involving coordination, conflict, and multiple equilibria.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:
1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question would be compulsory, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks (10x2=20).
4. Rest of the paper shall contain 4 units. Each Unit shall have two questions and the candidates shall be required to attempt on question from each Unit – 4 in all. Each question shall carry 15 marks (15 x 4 = 60).
5. Candidates are allowed to use Simple, Non-Scientific and Non-Programmable Calculator in the Examination Hall.
6. It is compulsory for the paper setter to mention the requirement of log, antilog and statistical tables.

UNIT-I

Linear Programming and Data Envelopment Analysis: Frontier Estimation using Linear Programming; measurement of economic, technical, revenue and profit efficiency scores. The Malmquist Productivity index.

Non-Linear Programming: Non-linear Programming and Kuhn-Tucker conditions. War time rationing and Peak load pricing issues.

UNIT-II

Dynamic Optimization: A Brief introduction to the calculus of variations (Euler's equation), endpoint (transversality) conditions. The control problem, Statement and types of control. Pontryagin's maximum principle, finite and infinite time horizon, interpretation of hamiltonian, costate variables, equation of motion; end point conditions. Constructing a phase diagram and its analysis.

UNIT-III

Input-Output Analysis: Static Open and Closed Models; Viability Conditions; Activity Analysis Interpretation of Input-Output Model; Aggregation Problem; Substitution Theorem; Cost and Price Determination; Dynamic Leontief-System. Social Accounting Matrix: An Introduction.

UNIT-IV

Game Theory: Two persons Zero Sum Game, Max-Min Strategy. Non-Zero Sum Game, Static Games in Complete Information, Strongly and Weakly Dominant Strategies, Dominant Strategy Equilibrium; Pure and Mixed Strategies; Concept of Nash Equilibrium–Examples such as Prisoner's Dilemma, Battle of Sex, Matching Pennies, Co-ordination Problem and Multiple Equilibria, etc., Focal Points, Dynamic Games in Complete Information, Sub-Games and Sub-Game Perfect Nash Equilibrium.

Recommended Readings:

Abel, A. & Bernanke, M, (1998). Macroeconomics. Massachusetts: Addison Wesley.

Allen, R. G. D. (1967). Macro Economic Theory. London: Macmillan.

Allen, R. G. D. (1974). Mathematical Economics. London: Macmillan

Baumol, W. J. (2000). Economic Theory and Operations Analysis (4th ed.). New Delhi: Prentice Hallof India.

Dorfman, R. , Samuelson, P. & Sallow, R. (1968) Linear Programming and Economic Analysis.Tokyo: McGraw Hill Kogakusha Ltd.

Dornbusch, R. & Fisher, S. (1981). Macroeconomics. New Delhi: McGraw Hill.

Hamberg, D. (1971). Models of Economic Growth. New York: Harper & Row.

Henderson, J. M. & Quandt, R. B. (1980). Micro Economic Theory: A Mathematical Approach (3^{rd}ed.). New York: John Wiley.

Lancaster, V. (1965). Mathematical Economics. R and McNally College Pub.

Madnani, G. M. K. (2003). Mathematical Economics. New Delhi: Oxford IBH Publishing Co. Pvt.Ltd.

Nehar, P. A. (1971). Economic Growth and Development: A Mathematical Introduction. New York:John Wiley

Silberberg, E. (1990). The Structure of Economics– A Mathematical Analysis. New York: McGraw Hill.

Varian, H. R. (1992). Microeconomic Analysis. New York: W. W. Norton.

Chand, M. & Anand, V. K. (1981). Economic Theory – A Mathematical Approach. Allahabad:

Kitab Mahal.
Chiang, A. C. (1984). Fundamental Methods of Mathematical Economics (3rd ed.). McGraw Hill.
Chung, J. W. (1993). Utility and Production: Theory and Application. London, New York: Basil Blackwell.
Green, H. A. J. (1971). Consumer Theory. Harmondsworth : Penguin.
Jha, R. (1991). Contemporary Macroeconomic Theory and Policy. New Delhi: Wiley Eastern Ltd.
Jones, H. G. (1976). An Introduction to the Modern Theory of Economic Growth. Tokyo : McGraw Hill - Kogakusha.
Leontief, W. W. (1966). Input-Output Economics. New York: Oxford University Press.
Nash, J. F. (1996). Essays on Game Theory. U.K: Cheltenham.
Quirk, J. & Saposnik, R. (1968). Introduction to General Equilibrium Theory and Welfare Economics.New York: McGraw Hill.
Sen, A. K. (1990). Growth Economics, Harmondsworth, Penguin.
Solow, Robert M. (2000). Growth Theory – An Exposition. Oxford: Oxford University Press.
Taha, H. S. (1997) Operations Research – An Introduction (6th ed). New Delhi: Prentice Hall of IndiaPvt. Ltd.,
Todaro, M. P. (1971). Development and Planning Models and Methods. Oxford: Oxford University Press

SEMESTER-IV

Level of Course	M.A. with Research		
	Course Code	**Course Name**	**Credits**
L-500	MAECORBC-RM-401 (B)	Research Publications and Ethics	4
	MAECODSC-402	Economics of Growth, Development and Planning-II	4
	Any *One*Paper from: MAECODSC/ DSE-403 or MAECODSE-404 (A) or MAECODSE-404 (B) or MAECODSE-404 (C) or MAECODSE-404 (D)	**Any *One* Paper from:** Indian Economy-II or Econometrics-II or Economics of Agriculture-II or Economics of Industry-II or Mathematical Economics-II	4
	MAECORBC-DST-405 (P2)	Dissertation	8

MAECORBC-RM-401 (B): RESEARCH PUBLICATIONS AND ETHICS

Topologyofthecode	**Research Based Course**
LevelofthCourse	**500**
CourseDuration/LTP(Lectures, Tutorials, Practical)	**L=60 hours(4 hours aweekfor15weeks=01 semester), T=00 hours, P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment20 marks**
Time	**3hours**

Objectives:

This course aims to develop an understanding of ethical issues related to research and publications students will acquire skills to write research papers/thesis writing following publication ethics. Indexing, citation databases and open access publications will be introduced in this course.

Pedagogy of the Course Work:

The time–tested method of imparting verbal instructions through lectures would be used. Examples, in so far as possible, would be selected from everyday life/experience. The feedback would be sought through the students' written assignments, tutorials and seminars. The students would be evaluated at the end of the academic year through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Course outcomes:

After completing this course, students will gain a clear understanding of ethical principles in academics and research, including intellectual honesty and research integrity. They will be able to identify and avoid scientific and publication misconduct such as fabrication, falsification, and plagiarism, and practice responsible academic writing and referencing. The course will equip students with skills to follow ethical publication practices, handle complaints and appeals, and use open-access and journal selection tools effectively. Students will also develop awareness of copyright issues and the ethical use of plagiarism detection software in scholarly work.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:

1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question would be **compulsory** and shall be a short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks (10×2 = 20).
4. The rest of the paper shall contain four units. Each unit shall have two questions and the candidates shall be required to attempt one question from each unit – 4 in all. Each question shall carry 15 marks (15×4 = 60).
5. Candidates are allowed to use Simple, Non-Scientific and Non-Programmable Calculator in the Examination Hall.
6. It is compulsory for the paper setter to mention the requirement of log, antilog and statistical tables.

UNIT – I

lntroduction to Ethics: Nature and Scope. Intellectual honesty and research integrity. Scientific misconducts: Falsification, Fabrication and Plagiarism (FFP).

UNIT-II

Importance of Good Academic Writing. Difference between Academic Writing and other writings. Referencing style and bibliography.

UNIT-III

Publication ethics: definition and importance. Publication misconduct: definition, concept, problems that lead to unethical behaviour. Identification of publication misconduct, complaints and appeals.

UNIT-IV

Online Publication Tools Open access publications. Journal finder/ journal suggestion tools. Issues of Copyright: Use of plagiarism software.

Recommended Readings:

Bird,A.(2006). Philosophy of Science. Routledge.

Israei, M. and I. Hay (2006). Research Ethics in Social Sciences. Sage.

Macintyre, Alasdair (1967) A Short History of Ethics. London.

P. Chaddah, (2018) Ethics in Competitive Research: Donotgets cooped; donot get plagiarized, ISBN: 978-9387480865

National Academy of Sciences, National Academy of Engineering and Institute of Medicine. (2009). On Being a Scientist: A Guide to Responsible Conduct in Research: Third Edition. National Academies Press.
Resnik, D. B. (2011). What Is ethics in research & why is it important? National Institute of Environmental Health Sciences, 1-10. Retrieved from https://www.niehs.nih.gov/research/resources/bioethics/whatis/index.cfm
Beall, J. (2012). Predatory publishers are corrupting open access. Nature, 489(7415). 179-179.
Indian National Science Academy (INSA), Ethics in Science Education, Research and Governance (2019). ISBN:978-81-939482-l-7. https//www. insaindia.res.in/pdf /Ethics

MAECODSC-402: ECONOMICS OF GROWTH, DEVELOPMENT AND PLANNING– II

Topology of the code	**Discipline Specific Core**
Level of the Course	**500**
CourseDuration/LTP(Lectures, Tutorials, Practical)	**L=60 hours(4 hours aweekfor15weeks=01 semester), T=00 hours, P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment20 marks**
Time	**3hours**

Objectives:

As a sequel to the post-second war developments, the study of Economic Development gained impetus because three-fourths of humanity was experiencing wretched conditions of existence. There was a pressing need in those countries for uplifting their economic conditions by restructuring their economies to acquire greater diversity, efficiency and equity, in consonance with their priorities. Since a variety of perspectives were available, the policy makers were eager to acquaint themselves with various policy options in their bid to re-construct their dilapidated economies. In addition, various international bodies were also keen to help and guide the laggards. Consequently, the study of development economics assumed greater significance. In recent times, the resurgence of marketism and greater focus on areas like education, health, sanitation, energy and environment, and infrastructure development, hitherto relegated to the background, have reopened some of the old debates besides opening up new areas of investigation.

Growth and Development economists are making earnest efforts at theorizing to break fresh grounds. Consequently, study of this discipline continues to be of prime importance. Modules incorporated in this paper are devoted to the theories of growth and development, importance of agriculture, and the rational and pattern of industrialization in developing countries. The other important issues in the context of development such as infrastructure – linkages, role of international trade, importance of economic policies and relevance of planning have been included in the modules of this paper.

Pedagogy of the Course Work:

The time–tested method of imparting verbal instructions through lectures would be used. Examples, in so far as possible, would be selected from everyday life/experience. The feedback would be sought through the students' written assignments, tutorials and seminars. The students would be evaluated at the end of the academic year through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Course outcomes:

On successful completion of this course, students will develop a rigorous understanding of the processes of economic growth, structural transformation, and income distribution in developing economies. They will be able to critically analyse agriculture–industry linkages, alternative growth strategies, and the role of trade, foreign aid, and investment in economic development. The course will enhance students' capacity to evaluate market and state roles, including market and government failures, planning mechanisms, and evolving development paradigms. Students will also acquire analytical skills to assess policy choices in the context of globalization and post-Washington consensus frameworks.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:

1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question would be compulsory, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks (10x2=20).
4. Rest of the paper shall contain 4 units. Each Unit shall have two questions and the candidates shall be required to attempt on question from each Unit – 4 in all. Each question shall carry 15 marks (15 x 4 = 60).
5. Candidates are allowed to use Simple, Non-Scientific and Non-Programmable Calculator in the Examination Hall.
6. It is compulsory for the paper setter to mention the requirement of log, antilog and statistical tables.

UNIT-I

Economic Growth and Structural Change: Structural Changes in the Composition of Gross Domestic Product and Occupational Structure.

Exploring the Relationship between Economic Development and Income Distribution: Kuznets' inverted U–Shaped Curve and Augmented Kuznets' Curve.

UNIT-II

Agriculture-Industry Interface: The Models of Lewis, Fei and Ranis and Todaro.

The Balanced Growth Doctrine (Rosenstein Rodan), Unbalanced Growth (Hirschman's version), and the Reconciliation of the two.

UNIT-III

International Trade and Economic Development (Three Stories about the Trade and Development Relationship).

Foreign aid and Economic Development; Role of Foreign Direct Investment (FDI) and Multi-National Corporations (MNCs) in the Emerging Scenario.

Investment Criteria: Investment Criteria; Choice of Technique.

UNIT-IV

Market and State: An Overview of the Economic Functions of the Market and State. Efficiency of the Competitive Market, Market Failure, Government Failure, on the Choice of Economic System, Market Failure in a Dynamic Economy.

Planning and Market: Planning by direction, Planning by market, Planning in backward areas.

From Washington to post-Washington consensus.

Recommended Readings:

Agarwala, A. N., & Singh, S. P. (Eds.). (1963). *Economics of underdevelopment: A series of articles and papers*. London: Oxford University Press.

Cypher, J. (2014). *The process of economic development*. London: Routledge.

Lucas Jr, R. E. (1988). On the mechanics of economic development. *Journal of Monetary Economics*, 22(1), 3-42.

Meier, G. M., & Rauch, J. E. (Eds.). (1995). *Leading issues in economic development (Vol. 6)*. New York: Oxford University Press.

Ray, D. (1998). *Development economics*. New Jersey: Princeton University Press.

Romer, P. M. (1986). Increasing returns and long-run growth. *Journal of Political Economy*, 94(5), 1002-1037.

Romer, P. M. (1990). Endogenous technological change. *Journal of Political Economy*, 98(5, Part 2), S71-S102.

Schultz, T. W. (1982). *Investing in people: The economics of population quality*. USA: University of California Press.

Sen, A. (1999). *Development as freedom*. New Delhi : Oxford University Press.

Thirlwall, A.P. (Latest edition). *Economics of development: Theory and evidence*. New York: Palgrave Macmillan.

Todaro, M. P., & Stephen, C. S. (Latest edition). *Economic Development*. New York: Addison Wasley Logman.

MAECODSC/DSE-403: INDIAN ECONOMY –II

Topologyofthecode	**Discipline Specific Core**
LeveloftheCourse	**500**
CourseDuration/LTP(Lectures, Tutorials, Practical)	**L=60 hours(4 hours aweekfor15weeks=01 semester), T=00 hours, P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment20 marks**
Time	**3hours**

Objectives:

The objective of this paper is to acquaint the students with the performance of different sectors of the Indian economy and the policy framework governing them. This will provide them an insight into the past, present and future functioning of the Indian economy and strengthen their analytical capability. The time-tested method of imparting verbal instructions through lectures would be used. Examples, in so far as possible, would be selected from everyday life/experience. The feedback would be sought through the students writing assignments, tutorials and seminars.

The students would be evaluated at the end of each semester through subjective type questions/answers(both short and essay type). The script would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Pedagogy of the Course Work:

The time–tested method of imparting verbal instructions through lectures would be used. Examples, in so far as possible, would be selected from everyday life/experience. The feedback would be sought through the students' written assignments, tutorials and seminars. The students would be evaluated at the end of the academic year through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Course outcomes:

Upon completion of this course, students will be able to analyse the growth and structural transformation of India's service sector and infrastructure development in the post-reform era, including policies on SEZs, competition, and consumer protection. They will understand the structure and reforms of the financial system, money and capital markets, and the regulatory role of SEBI. Students will be able to evaluate trends in public finance, fiscal imbalances, public debt, and the implementation of fiscal responsibility measures. The course will also enable students to assess India's external sector performance, trade policy reforms, foreign investment trends, and engagement with the World Trade Organization.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:

1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question would be compulsory, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks (10x2=20).
4. Rest of the paper shall contain 4 units. Each Unit shall have two questions and the candidates shall be required to attempt on question from each Unit – 4 in all. Each question shall carry 15 marks (15 x 4 = 60).

5. Candidates are allowed to use Simple, Non-Scientific and Non-Programmable Calculator in the Examination Hall.
6. It is compulsory for the paper setter to mention the requirement of log, antilog and statistical tables.

UNIT – I

Service Sector and Infrastructure Development in Post-reform Era:
Service Sector in India – Growth, Pattern and Future Prospects.
Infrastructure Development - Need, Performance and Government Strategy on Infrastructure Development. Special Economic Zones (SEZs): Policy, Features and Problems.
Changes in Public Policy with Special Reference to Competition Policy and Consumer Protection Act.

UNIT – II

Financial Sector in the Post-reform Era:
Structure and Features of the Financial System in India. Financial Sector Reforms.
Indian Money and Capital Markets — Structure, Features and Reforms.
Role of Securities Exchange Board of India (SEBI)

UNIT – III

Public Finance in the Post-reform Era:
Public Expenditure in India: Trends and Issues. Fiscal Imbalances and Public Debt Sustainability in theIndian Economy.
Fiscal Responsibility and Budget Management Act, 2003. Approach and Recommendations of Recent Finance commission.

UNIT- IV

External Sector the Post-reform Era:
Trade Policy during Post Reform Period. Liberalised Exchange Rate Management System. Foreign DirectInvestment – Trends and Issues. India and the World Trade Organisation.

Recommended Readings:
Books:
Acharya, S. & Rakesh, M. (2011). India's Economic Performance and Challenges (Essay's in Honour ofMontek S Ahluwalia). New Delhi: Oxford University Press.

Ahluwalia, I.J. & Little, IMD. (2008). India's Economic Reforms & Development (Essays in Honour ofManmohan Singh). New Delhi: Oxford University Press.

Dhar, P.K. (Latest ed.) Indian Economy- Its Growing Dimensions. Kalyani Publisher.

Datt, G. & A. Mahajan. (Latest Edition). Indian Economy. New Delhi: S.Chand .

Fernando, A.C. (2016). Indian Economy. India: Pearson.

Kapila, U. (Latest Edition) Indian Economy since Independence. New Delhi: Academic Foundation.

Kapila, U. (Latest Edition) Indian Economy: Performance and Policies. New Delhi: AcademicFoundation.

Krueger, A.O. (2002). Economic Policy Reforms and the Indian Economy. New Delhi: Oxford University Press.

Mishra, S.K. and Puri, V.K. (Latest Edition) Indian Economy. Mumbai: Himalya Publisher House.Economic and Political Weekly, Various Issues.

Economic Survey (Various Issues), Government of India. New Delhi: Oxford University Press.

"Essays from EPW-Quarter Century of Liberalisation in India" (2018).Oxford University Press.

Bhagwati, J.& Panagariya, A. (2012). India's Tryst with Destiny. India: Harper Collins.

Drèze, J., & Sen, A. (2013). An uncertain Glory-India & its Contradictions. New Delhi: Penguin Books.

Singh, M. (2018). Changing India Vol. 1-5. Oxford University Press.

Research Papers:

Acharyya, Rajat. (1994). Liberalised Exchange Rate Management system and Devaluation in India: TradeBalance Effect, Journal of Economic Integration, vol. 9, No. 4, December.

Banga, Rashmi. (2006). Critical Issues in India's Service led Growth, INRM policy Brief (2), AsianDevelopment Bank.

Gopalam, R. & M. C. Singh. (2015). Services Growth in India, Economic & Political Weekly, L(41).

Kaur, B., Mukherjee, A., Kumar, N. & Ekka, A. P. (2014). Debt Sustainability at the State Level in India,RBI working paper series, Department of Economic and Policy Research, RBI

Kaur, Balbir. (2012). Threshold Level of Debt and Public Debt sustainability: the Indian Experience, RBIaccessional paper, 33 (1 &2).

Meena, R.S. (2011). Financial Reformns: Realities and Myths.Ijmt, 19(1), 44-45.

Nagaraj, R. (2003). Foreign Direct Investment in India in the 1990's - Trend and Issues, Economic & Political Weekly, Vol.38, No.17, 26April.

Patil, R. H. (2006). Current state of the Indian capital market. Economic & Political Weekly, 41(11), 1001–1011.

Patil, R.H. (2010) Financial sector Reforms: Realities and myths, Economic & Political Weekly, XLV, (19), 8 May.

Pattnaik, R.K. et.al. (2003). Exchange Rate Management: The Indian Experience, Economic & Political Weekly, Vol. 38, no.2, 31 May.

Sahoo, P. (2015). Time to review the Special Economic Zone Act, Economic & Political Weekly, L (14).

Sally, Razeen. (2011). Indian Trade Policy after the crisis, ECIPE occasional papers, (4).

MAECODSE-404 (A): ECONOMETRICS – II

Topologyofthecode	**Discipline Specific Elective**
Level	**500**
CourseDuration/LTP(Lectures, Tutorials, Practical)	**L=60 hours(4 hours aweekfor15weeks=01 semester), T=00 hours, P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment20 marks**
Time	**3hours**

Objectives:

Application of economic theory need a reasonable understanding of economic relationships and relevant statistical methods. The econometric theory thus becomes a very powerful tool for understanding of applied economic relationships and for meaningful research in economics. This paper accordingly is devoted to equip the students with basic theory of econometrics and relevant applications of the methods. The topics covered in the course include various problems faced in estimation of both single equations and simultaneous equations models.

Besides the time-tested method of imparting verbal instructions through lectures, description and derivation of econometric models would be demonstrated by extensive use of blackboard. Examples, in so far as possible, would be selected from everyday life/experience. The feedback would be sought through the students writing assignments, tutorials and seminars.

The students would be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Pedagogy of the Course Work:

The course relies on a combination of lectures, solving problems, and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. The students would be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The script would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Course outcomes:

Upon completion of this course, students will be able to analyze and estimate simultaneous equation models (SEMs), addressing issues of identification, simultaneity bias, and instrumental variable techniques. They will understand the use of dummy variables in regression analysis, including Chow tests, and estimate qualitative response models such as Linear Probability, Logit, Probit, and Tobit models. Students will gain expertise in panel data analysis, including fixed and random effects models, distributed lag, and autoregressive models, along with techniques like the Koyck and Almon approaches. The course will also equip students with advanced time series analysis skills, including stationarity tests, unit roots, cointegration, Granger causality, and forecasting using AR, MA, ARIMA, Box–Jenkins methodology, and Vector Autoregression (VAR) models.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:

1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question would be compulsory, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to

attempt any 10 short answer type questions. Each short answer type question would carry 2 marks (10x2=20).

4. Rest of the paper shall contain 4 units. Each Unit shall have two questions and the candidates shall be required to attempt on question from each Unit – 4 in all. Each question shall carry 15 marks (15 x 4 = 60).

5. Candidates are allowed to use Simple, Non-Scientific and Non-Programmable Calculator in the Examination Hall.

6. It is compulsory for the paper setter to mention the requirement of log, antilog and statistical tables.

UNIT-I

Simultaneous Equations Models (SEM):

Nature of SEM's, Simultaneous Equation Bias, Identifications Problem. Rank and Order Conditions. Testing Identification of Economic Models. Estimation of SEM'S: ILS, 2SLS, 3SLS. Method of Instrumental Variables

UNIT-II

Dummy variables: Dummy Explanatory Variables and Their Use to Test Equality to Two Regressions; Chow Test and Dummy Variable Approach. Qualitative Response Regression Models: Nature of Qualitative Response models. Description and Estimation of Linear Probability Model and Problems. Logit, Probit and Tobit Models.

UNIT-III

Panel Data Models: Introduction: Advantages and Issues Involved in Utilizing Panel Data.

Simple Panel Data Models: Fixed Effects (LSDV) and Random Effects Approach. Distributed Lag and Autoregressive Models: Introduction and Nature of Distributed Lag (DL) & Auto Regressive (AR) Model. Koyck Approach for Estimation of DL & AR Models Rationalization of Koyck Approach by Adaptive Expectation and Partial Adjustment Hypothesis; Almon's Polynomial Approach.

UNIT-IV

Time Series Analysis: Testing Causality in Economics: Granger Causality Test. Stationary, Test of Stationary, Spurious Regression, Unit Roots, Dickey-Fuller Test, Cointegration, Engle Granger Test. Forecasting: AR, MA and ARIMA processes, Box Jenkins Methodology. Vector Auto Regression (VAR) Model, Introduction, Formulation, Estimation and Problems.

Recommended Readings:

Gujarati Damodar (2003). Basic Econometrics, (4th ed.), McGraw Hill, New York.

James H Stock, Mark W. Watson. Introduction to Econometrics (2011). (3rd ed.) Pearson Prentice Pvt.Ltd, Delhi.

Kementa, J. (1971). Elements of Econometrics, (2nd ed.), Macmillan, New York.

Madala, G. S. (2007). Introduction to Econometrics, (3rd ed.), John Wiley & Sons Ltd., England.

Johnston, J. & Dinarido. (2007). Econometrics (4th ed.).New York: McGraw Hill.

S. Hsiao Cheng (2003). Analysis of Panel Data, (2nd ed.).Cambridge : Cambridge University Press

MAECODSE-404 (B): ECONOMICS OF AGRICULTURE-II

Topology of the code	**Discipline Specific Elective**
Level	**500**
Course Duration/LTP (Lectures, Tutorials, Practical)	**L=60 hours(4 hours aweekfor15weeks=01 semester), T=00 hours, P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment20 marks**
Time	**3hours**

Objective:

The focus of this course is to introduce students to issues associated with agricultural finance, agricultural markets, food security and international trade in agricultural commodities, and also to expose students to the recent developments in Indian agriculture. The objective is to equip students to analysis and critically assess issues, policies and programmes in these areas.

Pedagogy of the Course Work:

The course relies on a combination of lectures, solving problems, and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. The students would be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The script would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Course outcomes:

This course provides students with a comprehensive understanding of agricultural economics, covering the role of agriculture in economic development, farm production, and inter-sector linkages. Students will analyse agricultural finance, credit markets, risk management, and policies, as well as the structure, efficiency, and stabilization of agricultural markets. The course examines theories and models of agricultural transformation, including Indian and international experiences, and equips students to evaluate demand-supply dynamics, trade, and food security issues. Overall, it enables students to assess agricultural development strategies, emerging challenges, and policy interventions for sustainable growth.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:

1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question would be compulsory, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks (10x2=20).
4. Rest of the paper shall contain 4 units. Each Unit shall have two questions and the candidates shall be required to attempt on question from each Unit – 4 in all. Each question shall carry 15 marks (15 x 4 = 60).
5. Candidates are allowed to use Simple, Non-Scientific and Non-Programmable Calculator in the Examination Hall.
6. It is compulsory for the paper setter to mention the requirement of log, antilog and statistical tables.

UNIT-I

Agricultural Finance: Financial Characteristics of Agriculture, Imperfect Information [Agency Relationship, Adverse Selection and Moral Hazard] and Enforcement Problems in Credit Markets; Segmented Rural Credit Markets, Resolving Information and Incentive Problems; Managing Borrowers' Credit Risks; Agribusiness and Trade Financing. Rationale for Government Intervention in Rural Credit Markets.

UNIT-II

Agricultural Markets: Nature, Types and Problems of Agricultural Commodity Markets; Marketing Margins and Marketing Margin Determinants; Evaluation of Marketing Efficiency. Horizontal Structure of Agricultural Marketing, Vertical Market Relationships.

Institutional and Organizations in Agricultural Markets: Farmers Cooperatives, Marketing Boardsand Trading Enterprises.

Instability in Agricultural Markets: Price Stabilization and Social Benefits, Storage and Market Stabilization, Role of Commodity Futures and Options in Stabilization.

UNIT-III

Food Security: Concept, Threat, Indicators and Mechanism to Food Security; Food Assistance Programmes [both Domestic and International].

Agriculture and International Trade: Main Features of International Trade in Agricultural Commodities, Domestic Agricultural and Trade Policies of Developed Countries and Their Impact on World Trade, Agriculture in GATT Negotiations and WTO.

UNIT-IV

Indian Agriculture: Structure and Functioning of Rural Credit Market in India.

Indian Agricultural Development Strategy Since Independence. Opportunities and Challenges [Expectations, Performance, and Emerging Constraints and Potentials] for Indian Agriculture in the Post Reform Period. National Agricultural Policy (Latest).

Recommended Readings:

Besley, Timothy (1994). "How Do Market Failures Justify Interventions in Rural Credit Markets?" TheWorld Bank Research Observer, Vol. 9, No.1

Bruce L. Gardener and Gordon C. Rausser (2002). (Ed.), Handbook of Agricultural Economics Vol., 2A

Agricultural and Its External Linkages, Amersterdam, Elsevier Science B.V.

Bruce L. Gardener and Gordon C. Rausser (2002). (Ed.), Handbook of Agricultural Economics Vol., 2B

Agricultural and Policy, Amersterdam, Elsevier Science B.V.

Bruce L. Gardener and Gordon C. Rausser (2002). (Eds), Handbook of Agricultural Economics Vol.1B,

Marketing Distribution and Consumption, Amersterdam, Elsevier Science B.V.

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Bruce L. Gardener and Gordon C. Rausser (2002). (Eds.), Handbook of Agricultural Economics Vol.1A

Agricultural Production, Amersterdam, Elsevier Science B.V.

Ghatak Subrata and Ken Ingersent (1984). Agriculture and Economic Development. London: Harvestor Press Ltd.
Hoff Karla & Joseph E. Stiglitz (1990). "Introduction: Imperfect Information and Rural Credit Markets – Puzzles and Policy Perspective" The World Bank Economic Review, Vol.4, No.3
Government of India Five Year Plan, (Latest), Planning Commission. New Delhi. Government of India,
Government of India National Agricultural Policy, Ministry of Agriculture, New Delhi. (Latest). Government of India.
World Bank (2006). Rejuvenating Indian Agriculture, Washington, World Bank.

MAECODSE-404 (C): ECONOMICS OF INDUSTRY-II

Topologyofthecode	**Discipline Specific Elective**
Level	**500**
CourseDuration/LTP(Lectures, Tutorials, Practical)	**L=60 hours(4 hours aweekfor15weeks=01 semester), T=00 hours, P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment20 marks**
Time	**3hours**

Objectives:
In the contemporary world with globalization and liberalization more and more attention is being given to industry. Since industry performance critically depends on firms' behaviour allowing equilibrium outcome, the course intends to provide a rigorous knowledge of different long-run equilibrium outcome
of firms under different conditions from the point of view of public policy. The students are also equipped to deal with debates involved in the industrial development in a cogent and analytical manner, particularly in the Indian context. However, it should be noted that Game Theoretic approach to any topic/problem is outside the scope of the Present syllabus.

Pedagogy of the Course Work:
The course relies on a combination of lectures, solving problems, and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. The students would be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The script would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Course outcomes:
This course provides students with a comprehensive understanding of industrial organization, market structures, and firm behaviour. Students will analyse profit and non-profit maximization hypotheses, monopoly and oligopoly power, strategic conduct under interdependence, and the role of innovation in competitive outcomes. The course covers mergers, takeovers, pricing strategies, price discrimination, and efficiency implications, along with the evaluation of market performance and public policy interventions. By integrating theoretical models with empirical and policy perspectives, students will be equipped to critically assess firm strategy, market efficiency, and the evolution of regulatory frameworks, including the Indian context.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:

1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question would be compulsory, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks (10x2=20).
4. Rest of the paper shall contain 4 units. Each Unit shall have two questions and the candidates shall be required to attempt on question from each Unit – 4 in all. Each question shall carry 15 marks (15 x 4 = 60).
5. Candidates are allowed to use Simple, Non-Scientific and Non-Programmable Calculator in the Examination Hall.
6. It is compulsory for the paper setter to mention the requirement of log, antilog and statistical tables.

UNIT – I

Rivalry; Conditions Facilitating and Hindering Collusive Conduct. Potential Competition: Limit Price versus Strategy by Dominant Firm. Direct cost-based strategy: rising Rivals Costs; Indirect Strategies:
capacity and marketing. Rivalry with Efficiency and Product Differentiation

Relationship between Market Structure and Technological Progress: Economics of innovation; Arrow and Schumpeter. Timing of Innovation and Innovation as a strategic Conduct.

UNIT – II

Market Power and Efficiency Related Causes of Different Types of Merger S: Horizontal; Vertical and Conglomerate. Causes of different types of Takeovers. Evaluation of Merger Policy: US experience.

UNIT – III

Macro Economic Issues: Means Thesis on Administered Pricing by Firms; The Kinked Demand Curve and Full Cost Pricing; Transaction Costs and Price Rigidity.

Issues of Price Discrimination: Nature of Price Discrimination. Effects of Price Discrimination: welfare, Efficiency and Competition.

UNIT – IV

Public Policy towards Market Structure, Conduct and Performance. Optimality of Perfect Competition.
Costs of Monopoly: Theoretical Issues and Empirical Measurement of Social Welfare Loses.

Evolution of Govt. of Indian Policy towards Monopolistic and Restrictive Practices: Theoretical issues.

Recommended Readings:

Bain, Wildsmith, S. (1975). Managerial Theories of the Firm.
Devine, P. J. & Others. (1976). Introduction to Industrial Economics, George Allen & Unwin.
Hawkin, C. J. (1973). Theory of the Firm. Macmillan.
Hay, D. A. & Morris, D. J. (1991). Industrial Economics: Theory and Evidence. Oxford University Press.
J. S. (1968) Industrial Organizations (2nd ed.), University of California.
Martin Stephen (1989). Industrial Economics (Economic Analysis and Public Policy).
Scherer, F. M. & Ross Boston David (1990) Industrial Market Structure and Economic Performance. Houghton Miffilin Company.
Shepherd, W. G. (1990). The Economics of Industrial Organisation (3rd ed.). New Jersey: Prentice Hall Englewood Cliffs.

MAECODSE-404 (D): MATHEMATICAL ECONOMICS-II

Topology of the code	**Discipline Specific Elective**
Level	**500**
Course Duration/LTP(Lectures, Tutorials, Practical)	**L=60 hours(4 hours aweekfor15weeks=01 semester), T=00 hours, P=0**
Credits	**4**
Max.Marks	**100Marks=Theory80marks+InternalAssessment20 marks**
Time	**3hours**

Objectives:

This course is designed to equip students to understand the economic concepts and theories by using mathematical tools and techniques to refine the verbal logic. The use of calculus has permitted formulation of economic problems in multivariable mode and yield valuable insight about optimizing human behaviour. Modern algebraic tools allow convenient handling of simultaneous equations in the context of linear programming, game theory and input-output analysis.

Mathematical economics deals with various applications of mathematical tools and techniques in defining and developing economic relationships. This course has been accordingly designed to include various mathematical techniques/methods/models related to the different parts of economic theory like consumer theory, theory of production, pricing, trade cycles, growth models etc. The paper covers important aspects of microeconomics; macroeconomics and development theory and the various modules have been accordingly designed.

Pedagogy of the Course Work:

The course relies on a combination of lectures, solving problems, and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. The students would be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The script would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Course outcomes:

This course develops a strong understanding of microeconomic and macroeconomic theory, covering consumer behaviour, production, market equilibrium, and welfare economics. Students will learn to apply mathematical models to analyse decision-making under risk, intertemporal choices, and policy

interventions. The course also examines classical, neo-classical, and endogenous growth models, enabling students to evaluate economic growth, capital accumulation, and development. Overall, it equips students to integrate economic theories for analysing real-world phenomena and policy impacts.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:

1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question would be compulsory, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks (10x2=20).
4. Rest of the paper shall contain 4 units. Each Unit shall have two questions and the candidates shall be required to attempt on question from each Unit – 4 in all. Each question shall carry 15 marks (15 x 4 = 60).
5. Candidates are allowed to use Simple, Non-Scientific and Non-Programmable Calculator in the Examination Hall.
6. It is compulsory for the paper setter to mention the requirement of log, antilog and statistical tables.

UNIT-I

Theory of Consumer Behaviour: Linear Expenditure Function and Indirect Utility Function; Roy's Identity; Expenditure Function; Consumer's Surplus; Choice under Risk, Von Neumann Utility Function;

Intertemporal Utility Maximization.

Production Analysis: Homogeneous and Homothetic Production Functions; Elasticity of Substitution; Cobb-Douglas, CES and VES Production Functions; Estimation of economic, technical, revenue, and profit efficiency scores using production CD, CES and VES functions.

UNIT-II

Market Equilibrium: Existence and Uniqueness of Equilibrium; Stability of Equilibrium; Dynamic Equilibrium with Lagged Adjustment; Cobweb Model. Multi-Market Equilibrium, Walras's Law and the

Existence of General Equilibrium.

Welfare Economics: Mathematical explanation of Paretian concept of social welfare. Bergsonian concept of social welfare. Externalities and economic welfare. The Compensation Principle: Kaldor Hicks criterion, Kaldor-Hicks criterion and Scitovsky community indifference curves. Scitovsky double criterion. Samuelson's comparisons. Little criterion. Social welfare function. Arrow's theorem.

UNIT-III

Fundaments of Macro Economics: Mathematical formulation of Classical and Keynesian models. Intertemporal utility function and consumption decisions, investment multiplier with once for all and repeating doses, the Balanced Budget Multiplier.

Relative Effectiveness of Fiscal and Monetary Policy: effectiveness issues under different regimes of IS-LM-BP Models: A Mathematical Representation.

UNIT-IV

Growth Models: Harrod-Domar Growth Models. An Overview of Neo-Classical Growth Models and Golden Rule of Capital Accumulation. Growth Accounting; Models of Endogenous Growth.
Two Sector Models of Endogenous Growth: Meade's Model, Jorgenson's model of a dual economy. Uzawa-Lucas model. Model with reversed factor intensities.

Recommended Readings:

Abel, A. & Bernanke, M, (1998). Macroeconomics. Massachusetts: Addison Wesley.
Allen, R. G. D. (1967). Macro Economic Theory. London: Macmillan.
Allen, R. G. D. (1974). Mathematical Economics. London: Macmillan
Baumol, W. J. (2000). Economic Theory and Operations Analysis (4th ed.). New Delhi: Prentice Hall of India.
Dorfman, R. , Samuelson, P. & Sallow, R. (1968) Linear Programming and Economic Analysis. Tokyo : McGraw Hill Kogakusha Ltd.
Dornbusch, R. & Fisher, S. (1981). Macroeconomics. New Delhi: McGraw Hill.
Hamberg, D. (1971). Models of Economic Growth. New York: Harper & Row.
Henderson, J. M. & Quandt, R. B. (1980). Micro Economic Theory: A Mathematical Approach (3rd ed.). New York: John Wiley.
Lancaster, V. (1965). Mathematical Economics. R and McNally College Pub.
Madnani, G. M. K. (2003). Mathematical Economics. New Delhi: Oxford IBH Publishing Co. Pvt. Ltd.
Nehar, P. A. (1971). Economic Growth and Development: A Mathematical Introduction. New York:John Wiley
Silberberg, E. (1990). The Structure of Economics– A Mathematical Analysis. New York: McGraw Hill.
Varian, H. R. (1992). Microeconomic Analysis. New York: W. W. Norton.
Chand, M. & Anand, V. K. (1981). Economic Theory – A Mathematical Approach. Allahabad: Kitab Mahal.
Chiang, A. C. (1984). Fundamental Methods of Mathematical Economics (3rd ed.). McGraw Hill.
Chung, J. W. (1993). Utility and Production: Theory and Application. London, New York: Basil Blackwell.
Green, H. A. J. (1971). Consumer Theory. Harmondsworth : Penguin.
Jha, R. (1991). Contemporary Macroeconomic Theory and Policy. New Delhi: Wiley Eastern Ltd.
Jones, H. G. (1976). An Introduction to the Modern Theory of Economic Growth. Tokyo : McGraw Hill - Kogakusha.
Leontief, W. W. (1966). Input-Output Economics. New York: Oxford University Press.
Nash, J. F. (1996). Essays on Game Theory. U.K: Cheltenham.
Quirk, J. & Saposnik, R. (1968). Introduction to General Equilibrium Theory and Welfare Economics.New York: McGraw Hill.
Sen, A. K. (1990). Growth Economics, Harmondsworth, Penguin.
Solow, Robert M. (2000). Growth Theory – An Exposition. Oxford: Oxford University Press.
Taha, H. S. (1997) Operations Research – An Introduction (6th ed). New Delhi: Prentice Hall of India Pvt. Ltd.,
Todaro, M. P. (1971). Development and Planning Models and Methods. Oxford: Oxford University Press.